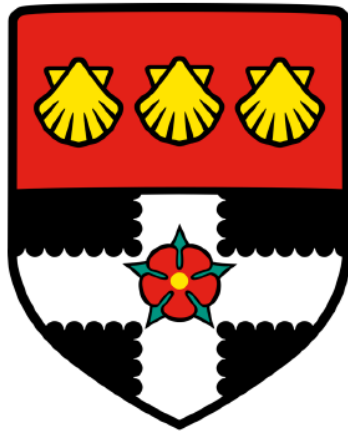


UNIVERSITY OF READING
DEPARTMENT OF METEOROLOGY



Improving Surface Gravity Wave Forecasts Using Data Assimilation

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Waves are a distinctive visual feature of the ocean surface, and their accurate forecasting is critical for ensuring safety in offshore operations. One company producing these forecasts is Fugro, who produce regional forecasts of quantities such as significant wave height for a client performing offshore operations in the Gulf of Mexico. The aim of this study is to use data assimilation to improve the accuracy of these forecasts.

Data assimilation (DA) refers to a range of statistical methods for integrating observations into a forecast, with the aim of making the forecast more accurate. The observations come from a buoy operated by the client, positioned near their offshore site in the Southern part of the Gulf. Observations of significant wave height (H_s) and two measures of wave period (T_{m01} , T_{m02}) are assimilated. One DA experiment (H_s DA) is performed where only significant wave height is assimilated, and another (all vars DA) is performed where significant wave height is assimilated alongside the two measures of wave period.

The DA method used borrows the analysis step of the ensemble transform Kalman filter (ETKF) method. The conventional ETKF consists of update steps in which observations from a given time are assimilated, alternating with forecast steps in which the forecast model is used to evolve the state forward to the next observation time. Since access to the forecast model is unavailable in this study, the forecast step cannot be performed and instead a single analysis step is used to update an entire 7-day forecast window simultaneously.

Since the errors in the forecast and observations are central in determining the working of DA, these errors are investigated prior to performing the assimilation. The forecast errors are seen to be closely related to the ensemble mean, and forecast error covariances are often confined to short distances, with some exceptions discussed. The observation errors are examined using a collocation study and found to be a strong linear function of the observations themselves. This motivates a scaled error approach in the DA, where observation errors are set at a constant percentage of the observed value.

Both H_s DA and all vars DA are found to reduce forecast RMSE by around 50% during the assimilation window, and improvements extend further in time in all vars DA. The ensemble spread after DA is examined in comparison to the spread before DA, and to the RMSE. Both DA experiments show a greater overall reduction in spread than in RMSE, thus increasing overconfidence in the ensembles. The errors in forecasts of the maximum significant wave height associated with a cold front on 12/03/2022 are examined, and it is found that at 1-2 days after assimilation, H_s DA improves forecasts of the height of the peak while all vars DA improves forecasts of its timing.

In conclusion, the DA method used here does improve multiple aspects of wave forecasts, with all vars DA generally outperforming H_s DA. It is hoped that this study will inform future wave DA work in the Gulf of Mexico, including a potential operational implementation by Fugro.

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List of Abbreviations

DA	Data Assimilation
EnKF	Ensemble Kalman Filter
ETKF	Ensemble Transform Kalman Filter
GEFS	Global Ensemble Forecast System
GoM	Gulf of Mexico
H_s	Significant Wave Height
LETKF	Local Ensemble Transform Kalman Filter
NOAA	National Oceanic and Atmospheric Administration
OI	Optimal Interpolation
OSSE	Observing System Simulation Experiment
PDAF	Parallel Data Assimilation Framework
T_{m01}	Mean Period
T_{m02}	Mean Zero-Crossing Period
WW3	WAVEWATCH III

Chapter 1

Introduction

1.1 The need for Reliable Wave Forecasting

Offshore operations form an important part of many industries and contribute to the economies of coastal countries round the world. Some examples are offshore wind farms in the renewable energy industry, and undersea cable maintenance in telecommunications. Ocean waves pose a significant risk to offshore operations as they can lead to the loss of life and damage to equipment (Adumene and Ikue-John, 2022; Ren et al., 2021). This means that accurate wave predictions are a critical safety issue, as offshore operations can only be carried out safely when the waves are sufficiently calm.

To provide these predictions, the MetOcean Forecasting division of Fugro produces wave forecasts for a client undertaking offshore operations in the Gulf of Mexico. It is important to understand the risk posed by waves in this region since increased cyclone intensity is leading to increased extreme wave heights in the Gulf of Mexico (Appendini et al., 2014). As well as needing to be accurate, it is important to know the uncertainty in these wave forecasts as the unnecessary pausing of operations is very costly, and knowledge of uncertainty can help to avoid and mitigate this cost.

To this end, Fugro produce ensemble forecasts, so that the ensemble spread can be used to infer uncertainty. As the production of ensemble forecasts is more costly than a single deterministic forecast, these ensemble forecasts are produced only when extreme conditions are expected, such as during winter storms. The rest of the time, a single deterministic forecast is produced instead. They produce non-ensemble forecasts year-round and ensemble forecasts are produced when extreme conditions are expected, such as during winter storms.

To account for uncertainty in the deterministic forecast, Fugro use a method called a site-specific alpha factor. This is based on the alpha-factor method developed by Det Norske Veritas (DNV, 2011) and further expanded upon by Gintautas and Sørensen (2016); Wilcken (2012); Wu and Gao (2021). This method seeks to apply corrections to the forecast by defining the so-called alpha factor

$$\alpha = \frac{H_{\max}}{H_{\max, \text{cor}}}, \quad (1.1)$$

where $H_{\max}$ is the wave height with a 10^{-4} probability of exceedance during a given window. $H_{\max, \text{cor}}$ is the wave height with the same exceedance probability during the same window, after the forecast is corrected for bias and uncertainty. Following Liu and Burcharth (1998), the Rayleigh distribution is used to model the probability distribution of maximum wave height over a window as follows:

$$\mathbb{P}(H \leq H_{\max}) = \left[1 - e^{\left(-2 \frac{H^2}{H_s^2} \right)} \right]^{\frac{t}{T_p}}. \quad (1.2)$$

Here H_s is the significant wave height during the window, t is the length of the window and T_p is the peak period during the window. $H_{\max}$ is then chosen so that $1 - \mathbb{P}(H \leq H_{\max}) = 10^{-4}$. $H_{\max, \text{cor}}$ is found by integrating the joint probability density function of H and H_s ,

$$\mathbb{P}(H, H_s) = \mathbb{P}(H|H_s)\mathbb{P}(H_s), \quad (1.3)$$

where $\mathbb{P}(H|H_s)$ is the Rayleigh distribution from equation 1.2 and $\mathbb{P}(H_s)$ is assumed to be a normal distribution with mean equal to the bias corrected forecast value of H_s and standard deviation equal to the forecast standard deviation. The bias and standard deviation were calculated from the timeseries of model output and observations spanning from September 2020 to May 2021. Integrating $\mathbb{P}(H, H_s)$ over all values of H_s gives a corrected distribution of maximum wave height during the window which can be used to define $H_{\max, \text{cor}}$ in the same way as $H_{\max}$.

Once the value of α is found, it is used to scale the H_s model output via $H_{s, \text{cor}} = \frac{H_s}{\alpha}$. So if the model predicts $H_s = 1.5\text{m}$ and $\alpha = 0.75$, the scaled forecast value is $H_{s, \text{cor}} = 2\text{m}$. Qualitatively speaking, α will always be less than 1, since forecast uncertainty acts to broaden the distribution of maximum wave height. This means that $H_{\max, \text{cor}}$ is always more than $H_{\max}$, as illustrated in figure 1.1.

Fugro has conducted research into the performance of the site-specific alpha factor method, but has not made the report publicly available. However, from personal correspondence with the Fugro metocean team (Nikolova, 2024), we learned that this method is overly cautious and systematically produces significant wave heights above those seen in the observational record. This conclusion was reached after examining the performance of the method in five strong storms between January and March 2022.

As well as producing these forecasts, Fugro has a buoy based near to the client which produces hourly observations of the sea state. Data assimilation (DA) refers to a range of techniques used to integrate observations into a forecast, with the aim of making the

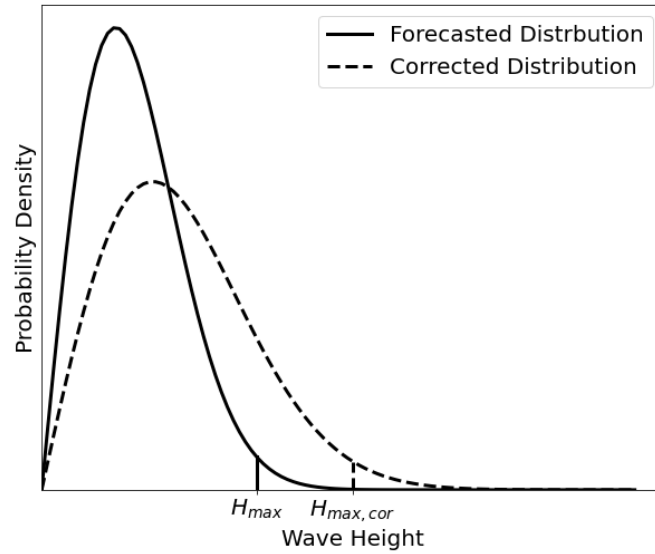


FIGURE 1.1: Schematic distributions of maximum wave height in a window, uncorrected (solid) and corrected for forecast bias and uncertainty (dashed).

forecast more accurate. In this project, the potential of DA to reduce uncertainty and error in these wave forecasts is explored. The aim is to develop a DA method which gives better forecasts than the alpha factor method currently in use.

1.2 Previous work on DA for Wave Forecasts

Several studies have been completed previously investigating the effectiveness of DA methods to improve wave forecasts. In this section, some of these studies and their findings are summarised, and the differences between this study and previous ones are discussed.

Aouf et al. (2006) assimilated spectral wave parameters into forecasts produced using the WAM model (Günther et al., 1992). The DA method they used is called optimal interpolation (OI) and produces an analysis field which is a weighted sum of the forecast pre-DA (called the background in DA terminology) and the difference between the background and the observations (called the innovations). The weights are determined by the estimated errors in the background state and observations, with a more detailed description given in section 2(b) of Aouf et al. (2006).

They did not assimilate real observations but rather pseudo-observations generated from model data, since the study was intended to investigate the impact of a proposed new observation system. This method is called a twin experiment or observing system simulation experiment (OSSE) and is commonly used for this purpose since it does not require any observational data (Zeng et al., 2020). In this study we also perform a twin experiment due to another advantage of the methodology: that the model data representing the truth is available at all times and locations, which allows for a more

thorough evaluation of forecast errors. More details on this aspect of the method are given in section 3.4.

Caires et al. (2018) investigated the effect on wave forecasts of using a DA method called the ensemble Kalman filter (EnKF) (Evensen, 2003) to adjust the wind fields used as forcing in a wave model. They used hourly observations of the 10-metre wind from buoys at six different locations. Their results are shown in figure 1.2. The RMSE with DA at two locations is plotted against forecast leadtime, and the forecast RMSE is compared to the RMSE at 0h leadtime with and without DA (referred to in the figure as hindcast and analysis respectively).

An improvement can be seen in forecasts of significant wave height and peak period, but it is relatively small compared to other studies examined here (Houghton et al., 2022, 2023) and short lived, not extending past 24hrs in any instance. This may be due to the indirect relationship between the assimilated wind data and the forecasted wave variables. In this study, observations of the forecasted wave variables are assimilated directly, and this is expected to lead to improved performance over the wind-based approach.

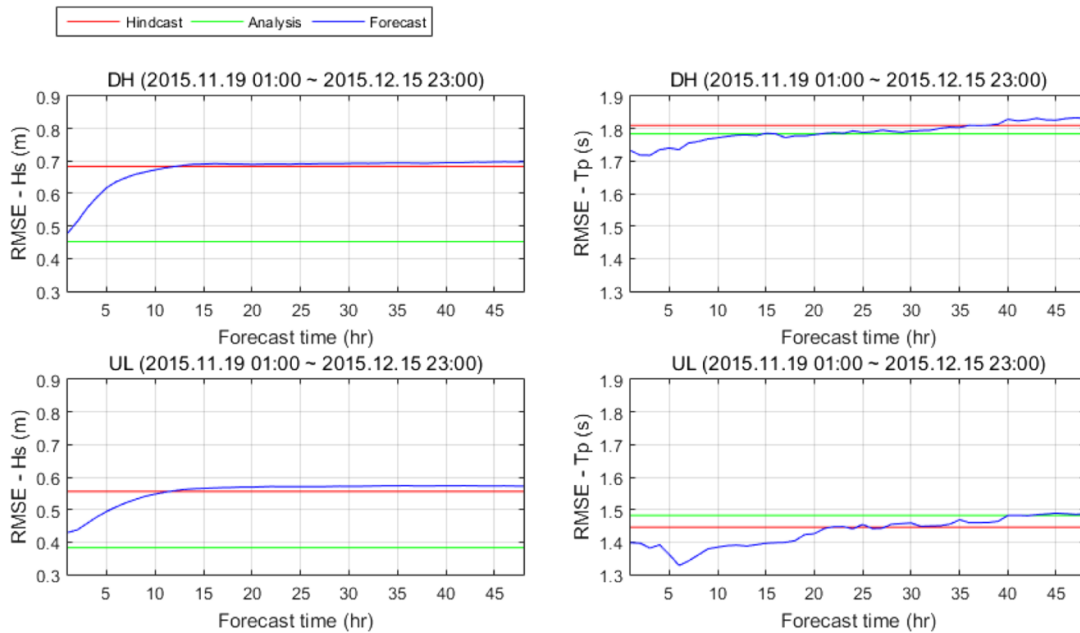


FIGURE 1.2: RMSE in significant wave height (left) and peak period (right) against forecast leadtime from two different observation locations. Figure 30 from Caires et al. (2018).

Another possible reason for the relatively poor performance in this case is the small number of observations used. Observations were available hourly from only six different locations, while operational DA systems commonly assimilate on the order of 10^7 observations at a time (Bannister, 2017). Our study is also limited in this regard, as data is only available from one buoy. A study which used many more observations is Houghton et al. (2022), in which an optimal interpolation (OI) method (Komen et al., 1994) was used to compare the effects of assimilating significant wave height and

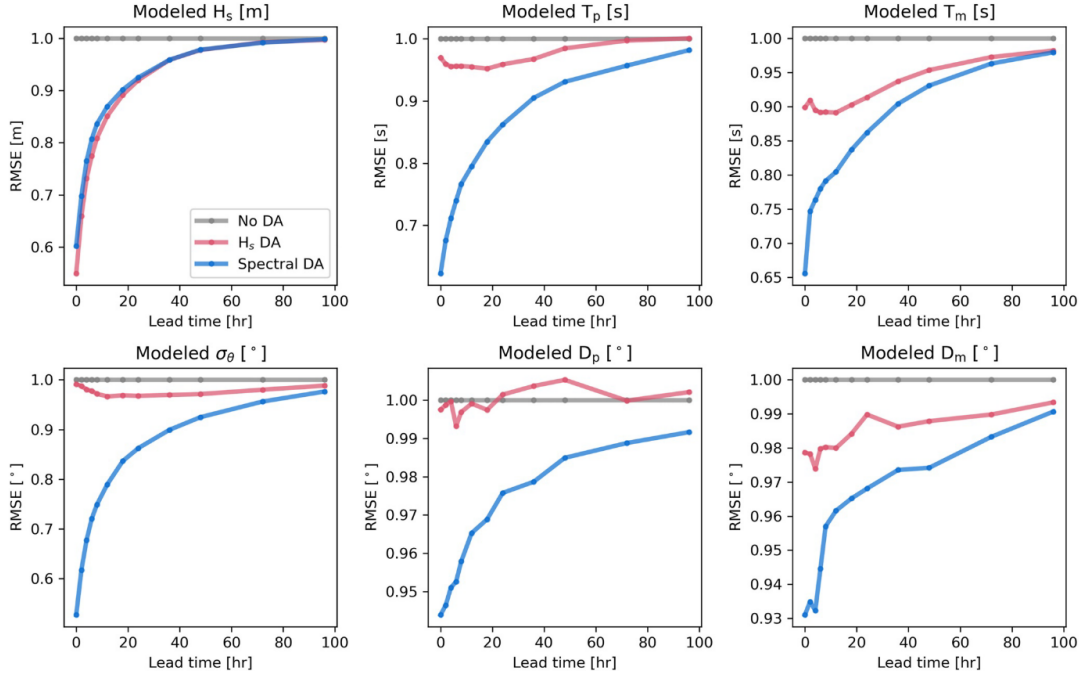


FIGURE 1.3: RMSE against forecast leadtime from spectral DA and significant wave height (H_s) DA, normalised by the RMSE without DA. Results shown for significant wave height H_s , peak period T_p , mean period T_m , directional spread σ_θ , peak direction D_p and mean direction D_m . Figure 3 from Houghton et al. (2022).

spectra. In this study, observations came from the global Sofar spotter buoy network (Raghukumar et al., 2019) which contains over 600 buoys. A plot of their results is shown in figure 1.3.

Figure 1.3 shows that both variables give very similar improvements in predictions of significant wave height, while spectral assimilation gives better results for other quantities such as period and direction. They also found that the spectral DA did particularly well in swell-dominated fields when compared to assimilating significant wave height. This shows that the type of observations assimilated can have a large effect on the performance of a DA system.

In this project H_s is assimilated along with two measures of wave period, T_{m01} and T_{m02} . These variables are related to different moments of the spectrum (see equations 2.8 - 2.11), so assimilating both T_{m01} and T_{m02} as well as significant wave height should have some of the benefits of spectral DA identified in Houghton et al. (2022). Since assimilating a full wave spectrum involves much more data than bulk variables such as significant wave height and period, the method used here is expected to incur less computational cost than assimilating the full spectrum.

To investigate the effect of the DA method used, Houghton et al. (2023) assimilated significant wave height observations from satellite altimeters and Sofar spotter buoys. They compared two different DA methods, OI and the ensemble-based local ensemble

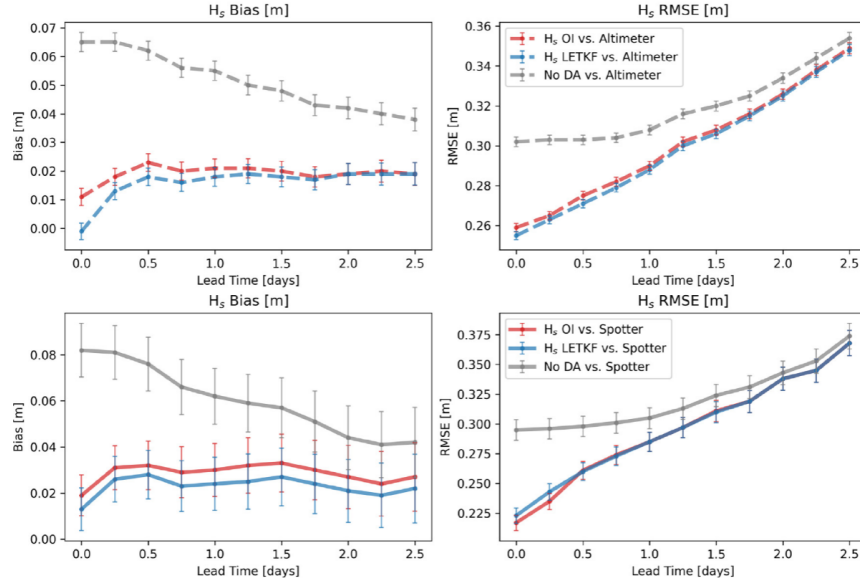


FIGURE 1.4: Forecast bias (left) and RMSE (right) from altimeter observations (top) and buoy observations (bottom), plotted against forecast leadtime. Figure 6 from Houghton et al. (2023).

transform Kalman filter (LETKF) (Hunt et al., 2007). Plots of their results are included in figure 1.4.

They found that both methods reduced forecast error and bias out to 2.5 days lead time, with LETKF having slightly greater forecast skill overall. In addition to this, they found that the LETKF gave more physically realistic increments due to its flow dependent background error covariance matrix. A plot showing increments from each method is included in figure 1.5. This shows that as well as the observations, the DA method used affects DA performance.

The method used in this project borrows the analysis step from the ensemble transform Kalman filter (ETKF) method (see section 2.2.2), which is similar to the LETKF except it does not use localisation and so allows observations to impact the state over long distances. Further discussion on the topic of localisation is included in section 2.2.3. This method was chosen over LETKF partly because it leads to a simpler DA implementation, and partly because the limited scope of the available observations means that localisation would stop the DA from making any change to the state across most of the domain. A major advantage of LETKF is its computational efficiency (Hunt et al., 2007), and borrowing techniques from the ETKF should also lead to an efficient method.

One other notable aspect of Houghton et al. (2023) is their diagnosis of the errors in the buoy observations. They used a colocation study to estimate the error standard deviation as a function of observed significant wave height, and found a strong linear relationship. As we are also using buoy observations, we expect that this linear relationship will also exist in our observations, so this method will be used to estimate the

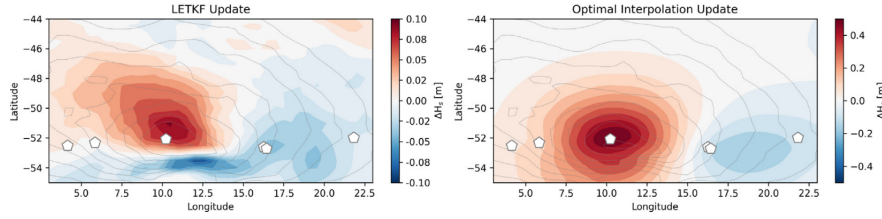


FIGURE 1.5: An example significant wave height increment from both LETKF and OI methods. Colours show value of increment, markers show the locations of observation buoys. Figure 3 from Houghton et al. (2023).

observation error variance. Further details are given in section 4.2.1.

In summary, observations of H_s , T_{m01} and T_{m02} are assimilated in order to capture the benefits of spectral DA identified in Houghton et al. (2022) while avoiding the computational cost. Our DA method borrows from the ETKF in the interests of simplicity and computational efficiency, and also to allow DA to update the state at all locations in the domain. The twin experiment methodology is used to allow for more thorough investigation of forecast errors. Following Houghton et al. (2023), a colocation study is used to estimate the observation error variance.

In this project, we apply DA to a specific forecasting context and client in the Gulf of Mexico, and so we are constrained by the observation data available in this context. As seen above, the studies with more observations saw better DA performance, and our results should be evaluated with this limitation in mind. Another limitation which none of the previous studies were subject to is that we are unable to run the wave model ourselves, which is a central step of most DA systems. We expect that the knowledge gained from these previous studies will allow us to get good results in spite of these limitations.

1.3 Aims of the Project

The main research question in this project is **“What kind of reduction in forecast error can be achieved by assimilating H_s , T_{m01} and T_{m02} into ocean wave forecasts?”**

To this end, we perform a twin experiment using a DA method which borrows from the ensemble transform Kalman filter (ETKF) to assimilate observations of the three variables. A twin experiment is a DA experiment in which fictional observations are generated and assimilated in place of real observations (see section 3.4 for details). We perform one experiment (all vars DA) in which all three aforementioned variables are assimilated, and one experiment (H_s DA) in which significant wave height alone is assimilated. The reasoning for this is that extreme wave heights are of particular interest to Fugro’s client due to their impact on operational safety, so we consider H_s to be the most important variable of the three. We examine the results of these two

experiments in comparison with each other, and with the forecast data before DA, to evaluate the performance of DA in both instances.

Various aspects of the results will be examined to gain a full understanding of the DA performance. These are:

- **Increments.** The changes made to the forecast by the DA system are called increments, and they can aid understanding of how DA is modifying the model state. We hypothesise that the increments will realistically reflect the information contained in the observations and act to move the forecast state closer to the truth. This is examined in section 5.1.
- **The root mean square error (RMSE) of the ensemble mean.** We hypothesise that the DA will reduce RMSE both at the buoy and away from the buoy to a lesser extent. This is examined in section 5.2.
- **Ensemble spread.** This is a commonly used indicator of uncertainty in ensemble forecasts, which can give an estimate of forecast error and allow a forecast to inform decisions with greater or lesser confidence. It is desirable then to have ensemble spread which is comparable with the forecast error so that this estimate is a good one. We hypothesise that DA will reduce ensemble spread, but not so much that it ceases to be a good measure of forecast error. This is examined in section 5.3.
- **Errors in maximum H_s .** Extremes of H_s are of particular interest in these forecasts as these are the events which cause danger and require the pausing of offshore operations. We hypothesise that the DA will reduce error in the timings and magnitudes of significant wave height maxima. This is examined in section 5.4.

The rest of this report is organised as follows. Chapter 2 covers the theoretical aspects of the study, which are required to understand the methods and results presented in later chapters. Chapter 3 describes the experimental design used in this study, including details of the wave model used and of the twin experiment and DA methodologies. Chapter 4 covers the nature of the background and observation errors, which are central to determining the form and magnitude of the corrections made by DA. Chapter 5 covers the results of the DA experiments, including examinations of the four aspects in the bullet points above. Chapter 6 summarises the methods and results of the study and makes recommendations for future work based on identified limitations.

Chapter 2

Theory

This chapter outlines the theoretical aspects of this study, covering both the linear wave theory used in the wave model and the theory behind data assimilation methods. The aim is to give the reader an overview of all the theory needed to understand the methods and results of the study.

2.1 Ocean Waves

Ocean waves are a distinctive visual feature of the ocean surface, and very important to offshore operations such as wind farms, oil and gas rigs and undersea cable maintenance. This section will describe some of the physical processes influencing ocean waves, as well as the theory of linear waves and wave energy spectra as modelled by the WAVEWATCH III (WW3) model used in this project (WW3DG, 2019).

2.1.1 Physical Processes

Ocean waves are an instance of surface gravity waves, meaning they propagate on the interface between two fluids of different densities. Their restoring force is gravity. The focus of this project is on waves with periods from 10s - 0.25s, which are generated by wind, although ocean waves can exist on a wide range of scales and be influenced by other factors such as surface tension and local air pressure variations (Holthuijsen, 2007, pp. 3-7).

There are multiple hypothesised physical mechanisms for the generation of waves by wind. Phillips (1957) shows that small-scale turbulent eddies induce small surface waves at frequencies matching the frequency of the turbulence. Miles (1957) describes how the surface height variations from these small waves lead to energy transfer from the wind to the waves, thus increasing wave height. A more detailed overall description of the various processes is given in Janssen (2004), chapter 3.

The wind almost always acts as a source of wave energy, so there has to be a process present to dissipate waves and act as an energy sink. These processes are poorly understood but Janssen (2004), section 4.9 describes multiple physical mechanisms which may contribute to wave dissipation. One important process affecting dissipation is

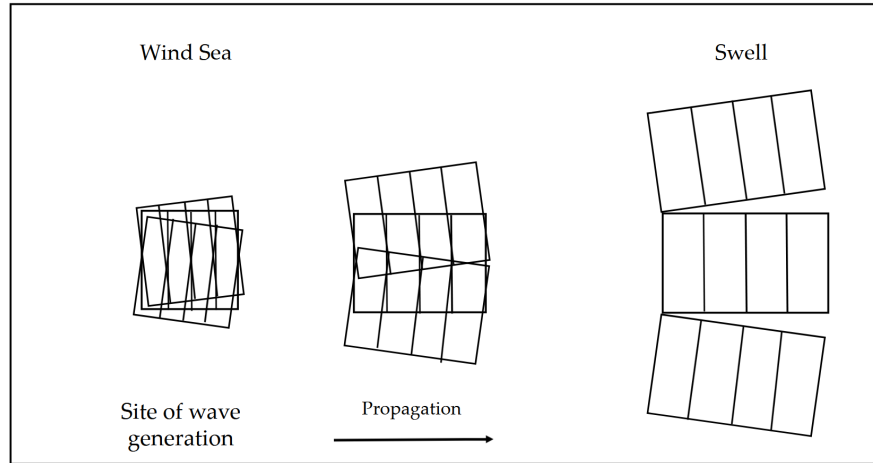


FIGURE 2.1: Schematic of the generation of wind and swell waves.

wave breaking, which occurs in near shorelines when the wave height gets too large compared to the water depth. It can also occur when wave height gets too large compared to wavelength. This steepness induced breaking is also known as white capping. Water's molecular viscosity can lead to the damping of shorter waves and larger waves can be damped by the surface eddies induced by wave breaking.

Ocean waves in deep water are dispersive, and waves with longer wavelengths travel faster than waves with shorter wavelengths (discussed quantitatively in section 2.1.2). This leads to a phenomenon known as swell, which is the name given to long waves that have travelled far from their source. Due to dispersion these waves have no short-wavelength component, so swell waves have longer wavelengths and periods than waves near the site of generation which are affected by local winds (known as wind sea). The other feature of swell waves is their regular appearance. While a wind sea has wave components in many different directions, by the time the waves have travelled away from the source these different components will spread out so that they do not overlap, and the resulting swell will be almost unidirectional. See figure 2.1 for a schematic.

2.1.2 Linear Wave Theory

This section will outline the linear wave theory relevant to this project, as derived in Vallis (2017), page 253. Linear wave theory is an approximation, and other higher-order theories of surface gravity waves exist, such as Stokes waves and cnoidal waves (Osborne, 2010). These higher-order theories will not be considered here since the WAVEWATCH III (WW3) model used in this project models the ocean surface as a linear superposition of simple linear waves.

Parameters describing linear waves include the wavenumber k , direction θ and frequency ω . The first two describe the wave vector $\mathbf{k}$, which has magnitude k and direction θ ,

$$\mathbf{k} = -k \sin \theta \mathbf{i} - k \cos \theta \mathbf{j}, \quad (2.1)$$

where $\mathbf{i}, \mathbf{j}$ are unit vectors pointing East and North respectively. Note that θ measures the wave direction in the meteorological convention i.e. the direction the wave is coming from. This is the same convention used in WW3, and means that $\theta = 0^\circ$ corresponds to a northerly wave, $\theta = 90^\circ$ corresponds to an easterly wave and so on.

Here, ω denotes the absolute frequency measured in a fixed frame of reference. If the medium is subject to currents, this will differ from the relative frequency σ measured in a frame of reference moving with the mean current due to the Doppler effect. This difference is quantified by the equation

$$\omega = \sigma + \mathbf{k} \cdot \mathbf{U}, \quad (2.2)$$

where $\mathbf{U}$ is the mean current vector. σ and k are related by the dispersion relation

$$\sigma^2 = gk \tanh kD, \quad (2.3)$$

where D is the water depth. The dispersion relation has two limiting forms:

- Deep water waves, where $kD \gg 1$, so $\tanh kD \approx 1$ and $\sigma^2 \approx gk$.
- Shallow water waves, where $kD \ll 1$, so $\tanh kD \approx kD$ and $\sigma^2 \approx gk^2D$.

The phase speed is then given by

$$c = \frac{\sigma}{k} \approx \begin{cases} \sqrt{\frac{g}{k}} & \text{in deep water,} \\ \sqrt{gD} & \text{in shallow water,} \end{cases} \quad (2.4)$$

and the group speed may be expressed as

$$c_g = \frac{\partial \sigma}{\partial k} = \frac{1}{2}c \left(1 + \frac{2kD}{\sinh 2kD} \right) \approx \begin{cases} \frac{1}{2}c & \text{in deep water,} \\ c & \text{in shallow water.} \end{cases} \quad (2.5)$$

The phase speed is the speed at which the wave peaks and troughs move, whereas the group speed is the speed at which wave groups move. The group speed is then the speed at which energy is transported by the wave. The wave travels in the direction of the wave vector $\mathbf{k}$ so the group and phase velocities are given by

$$\begin{aligned} \mathbf{c} &= -c \sin \theta \mathbf{i} - c \cos \theta \mathbf{j}, \\ \mathbf{c}_g &= -c_g \sin \theta \mathbf{i} - c_g \cos \theta \mathbf{j}. \end{aligned}$$

Scale analysis can now be used to see that ω and σ generally have similar values. The ratio between $\mathbf{k} \cdot \mathbf{U}$ and σ in equation 2.2 is

$$\frac{\mathbf{k} \cdot \mathbf{U}}{\sigma} \sim \frac{kU}{\sigma} = \frac{U}{c},$$

where $U = |\mathbf{U}|$ is the mean current speed. Since $c \gg U$ in general, this ratio is small and so $\omega \approx \sigma$.

2.1.3 Wave Spectra

Note that equations 2.2 - 2.5 are valid for linear waves of a single frequency and direction. Real ocean waves are more complicated than this, containing multiple frequencies propagating in multiple directions. The WAVEWATCH III (WW3) model used in this project models ocean waves as a superposition of these simple linear waves.

Surface waves with multiple frequencies and directions are represented using a wave spectrum $F = F(\sigma, \theta)$, which is also a function of position and time. F represents the sea surface height variance associated with a particular frequency σ and direction θ , so the total height variance is

$$\overline{\eta^2} = \iint F \, d\sigma \, d\theta. \quad (2.6)$$

This is related to the wave energy per unit surface area E via $E = \rho g \overline{\eta^2}$. A rigorous definition of F is given in Holthuijsen (2007), section 3.5.6. One can also define the direction-integrated spectrum $S(\sigma) = \int F(\sigma, \theta) \, d\theta$, which quantifies the surface height variance as a function of frequency. This can be integrated to calculate the energy contained in a specific frequency band $\Delta\sigma$:

$$\Delta E = \rho g \int_{\Delta\sigma} S(\sigma) \, d\sigma. \quad (2.7)$$

S is a helpful diagnostic from which information about the appearance of the waves can be inferred. A widely spread spectrum implies irregular waves with many different frequency components whereas a tight spectrum implies regular waves with one dominant frequency. Low frequency waves correspond to swell, while high frequency waves correspond to a wind sea. An example of a direction-integrated spectrum is plotted in figure 2.2. This spectrum comes from a point in the GoM that is far from any shorelines. Two peaks are apparent on the spectrum, one at about 0.3 Hz which corresponds to wind waves and one at about 0.2 Hz, which corresponds to swell. The wide spread of this spectrum indicates that the waves probably have an irregular appearance.

Here, we also define the wave spectral moments which are derived from the direction-integrated spectrum:

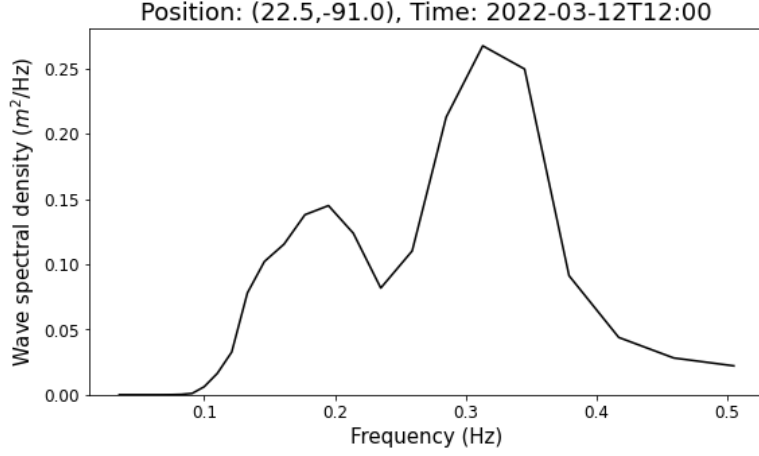


FIGURE 2.2: An example direction-integrated spectrum from a point in the gulf of Mexico. Calculated from WW3 spectral forecast data.

$$m_i = \int S(\sigma) \sigma^i d\sigma, \quad i = \dots, -1, 0, 1, \dots \quad (2.8)$$

Note that $m_0 = \overline{\eta^2}$ is the overall surface height variance. These moments give information about the statistical properties of the ocean, and can be used to derive other important wave variables from the spectrum. In this project we focus on the significant wave height H_s and two measures of period T_{m01} , T_{m02} . Holthuijsen (2007), section 4.2 (in which significant wave height is denoted H_{m0}) gives derivations of these variables from the spectral moments, reproduced below:

$$H_s = 4\sqrt{m_0}, \quad (2.9)$$

$$T_{m01} = \frac{m_0}{m_1}, \quad (2.10)$$

$$T_{m02} = \sqrt{\frac{m_0}{m_2}}. \quad (2.11)$$

T_{m01} is the mean wave period, that is the reciprocal of the mean frequency of the wave spectrum, and T_{m02} is the mean zero-crossing period, that is the average period between times when the wave-induced surface height displacement crosses zero. Finally we define the wave action

$$N(\sigma, \theta) = \frac{F(\sigma, \theta)}{\sigma}. \quad (2.12)$$

N is conserved in the absence of sources and sinks (Bretherton and Garrett, 1968), so it is very useful for wave modelling. WW3 works by evolving the wave action according to the effects of sources and sinks (see section 3.1).

2.2 Data Assimilation

Data assimilation (DA) refers to a group of statistical methods for incorporating observations into forecasts. It requires a model state and some observations, and the assumption that the observations are a function of some underlying true state of the system. The goal is to find the state that is most likely given the forecast and observation data, assuming some distributions for the errors in each.

2.2.1 Vector Representations and Cost Function

In DA, the state of a system must be represented with a vector so that methods of linear algebra can be applied to it. This section covers some of the details of this, as well as the treatment of observations and basic method by which the most likely state is found.

We take a vector $\mathbf{x} \in \mathbb{R}^n$ to represent the state of the system. This state is represented computationally in a model, and the form of $\mathbf{x}$ depends on the DA application. In some cases, $\mathbf{x}$ simply has one component for each variable of the system at each model gridpoint. Other applications use what is known as an augmented or 4D state vector, where $\mathbf{x}$ is expanded to contain components from times $t = 0, 1, \dots, T$ i.e.

$$\mathbf{x} = \begin{pmatrix} \mathbf{x}(t=0) \\ \mathbf{x}(t=1) \\ \vdots \\ \mathbf{x}(t=T) \end{pmatrix}.$$

An augmented state vector is used in this study. Given a first guess $\mathbf{x}_b$, we assume that the state of the system has a normal distribution $\mathbb{P}(\mathbf{x}) = N(\mathbf{x}_b, \mathbf{B})$. $\mathbf{x}_b$ here is called the background state, and is usually taken to be the state of the model before DA is applied. $\mathbf{B}$ is called the background error covariance matrix. Since the true background error covariance matrix is generally unknown, it must be estimated, and methods of achieving this are discussed in section 2.2.2.

We take a vector $\mathbf{y} \in \mathbb{R}^p$ to represent a set of observations, and assume $\mathbf{y}$ is related to $\mathbf{x}$ via an observation operator $\mathcal{H} : \mathbb{R}^n \rightarrow \mathbb{R}^p$ and an error term ε ,

$$\mathbf{y} = \mathcal{H}(\mathbf{x}_{\text{true}}) + \varepsilon,$$

where $\mathbf{x}_{\text{true}}$ is the unknown true state of the atmosphere. The error term ε is assumed to be normally distributed with mean zero and covariance $\mathbf{R}$ i.e. $\varepsilon \sim N(0, \mathbf{R})$. The matrix $\mathbf{R}$ is referred to as the observation error covariance matrix.

We call the vector spaces containing $\mathbf{x}$ and $\mathbf{y}$ the state space and observation space respectively. In most operational contexts, the dimensions of these spaces are very large: $\mathbf{x}$ has at least one component for each variable involved in the assimilation at each

model gridpoint, so typically $n \sim 10^8$. $\mathbf{y}$ has at least one component for each observed variable at each observation location, so $p \sim 10^7$. This means that considerable computational resources are required in most cases and computational efficiency is a key consideration in any operational DA implementation.

In our case, we have three variables on a 181×210 grid and each forecast window contains 169 timesteps, so the dimension of our state space is approximately 2×10^7 . We have only one observation location measuring the same three variables and we assimilate observations from 73 different timesteps, so the dimension of the observation space here is only 219. Note that these dimensions are smaller than those seen in operational contexts, so the DA problem is smaller with fewer degrees of freedom. This means that solving it computationally is practical on the hardware accessible in this study. Further details and reasoning for these numbers are given in section 3.5.

In our case $\mathcal{H}$ is linear, that is, there is a matrix $\mathbf{H} \in \mathbb{R}^{n \times p}$ such that $\mathcal{H}(\mathbf{x}) = \mathbf{H}\mathbf{x}$, and so the probability distribution of $\mathbf{y}$ given that the system is in state $\mathbf{x}$ is $\mathbb{P}(\mathbf{y}|\mathbf{x}) = N(\mathbf{H}\mathbf{x}, \mathbf{R})$. We can then express the distribution of $\mathbf{x}$ given some set of observations $\mathbf{y}$ using Bayes' theorem:

$$\mathbb{P}(\mathbf{x}|\mathbf{y}) = \mathbb{P}(\mathbf{y}|\mathbf{x}) \frac{\mathbb{P}(\mathbf{x})}{\mathbb{P}(\mathbf{y})}. \quad (2.13)$$

This allows $\mathbb{P}(\mathbf{x}|\mathbf{y})$ to be determined since $\mathbb{P}(\mathbf{y}|\mathbf{x})$ and $\mathbb{P}(\mathbf{x})$ are known, and $\mathbb{P}(\mathbf{y})$ is independent of $\mathbf{x}$ so it acts as a normalisation constant. Since all the distributions involved are normal, $\mathbb{P}(\mathbf{x}|\mathbf{y})$ will be normally distributed also,

$$\mathbb{P}(\mathbf{x}|\mathbf{y}) = A(\mathbf{y}) e^{-\frac{1}{2}(\|\mathbf{x} - \mathbf{x}_b\|_{\mathbf{B}^{-1}}^2 + \|\mathbf{y} - \mathbf{H}\mathbf{x}\|_{\mathbf{R}^{-1}}^2)}, \quad (2.14)$$

adopting the notation $\|\mathbf{a}\|_{\mathbf{M}}^2 = \mathbf{a}^\top \mathbf{M} \mathbf{a}$. $A(\mathbf{y})$ in this equation is a normalising constant whose value is chosen so that the integral of $\mathbb{P}(\mathbf{x}|\mathbf{y})$ over all values of $\mathbf{x}$ is equal to 1 for each $\mathbf{y}$.

We then seek the most likely value i.e. the mode of this distribution, since this will be a good estimate for the true state of the system given background state $\mathbf{x}_b$ and observations $\mathbf{y}$. Finding the mode of equation 2.14 is equivalent to minimising a cost function

$$J(\mathbf{x}) = \frac{1}{2} \|\mathbf{x} - \mathbf{x}_b\|_{\mathbf{B}^{-1}}^2 + \frac{1}{2} \|\mathbf{y} - \mathbf{H}\mathbf{x}\|_{\mathbf{R}^{-1}}^2. \quad (2.15)$$

The first term of 2.15 represents the deviation of $\mathbf{x}$ from the background state and the second term represents the deviation from the observations, with the overall cost function being a sum of these two weighted by the background and observation error matrices. This means that when the background error is large compared to the observational error, the observations will be given more weight and vice versa.

2.2.2 Kalman Filters

The specific DA method used in this study is based on the ensemble transform Kalman filter, which is part of a broad class of methods called Kalman filters. This section begins with a discussion of the overall properties of Kalman filters, followed by a more specific description of the method used in this study and the ensemble transform Kalman filter (ETKF).

The state that minimises the cost function in equation 2.15 is called the analysis $\mathbf{x}^a$ and may be expressed as

$$\mathbf{x}^a = \mathbf{x}_b + \mathbf{K}\mathbf{d}_b^o \quad (2.16)$$

where $\mathbf{d}_b^o = \mathbf{y} - \mathbf{H}\mathbf{x}_b$ is the innovation and $\mathbf{K} = \mathbf{B}\mathbf{H}^\top (\mathbf{H}\mathbf{B}\mathbf{H}^\top + \mathbf{R})^{-1}$ is the Kalman gain (derived in Asch et al. (2016), pp. 92-96).

Note that the increment $\mathbf{d}_b^a = \mathbf{x}^a - \mathbf{x}_b$ is given by $\mathbf{B} \left(\mathbf{H}^\top (\mathbf{H}\mathbf{B}\mathbf{H}^\top + \mathbf{R})^{-1} \mathbf{d}_b^o \right)$, so it always lies in the column space of $\mathbf{B}$. This means that $\mathbf{B}$ is crucial to determining how information from the observations is spread around the domain by the DA system. In particular, if $\mathbf{B}$ is rank deficient, then all possible increments lie within a strict subspace of the state space. This reduces the degrees of freedom in the corrections the DA system can make, which can limit the performance of the system.

One class of DA methods which use equation 2.16 are called Kalman filters, in which context it is called the analysis step. Kalman filters alternate this analysis step with a forecast step in which a forecast model is used to evolve the state forward to the next observation time. Kalman filters approximate $\mathbf{B}$ with a flow-dependent error covariance matrix $\mathbf{P}$ (equations 2.20 and 2.21) which is explicitly part of the DA output and which is evolved forward in time by the model along with the state. Kalman filters are a diverse class of methods, with many variants existing (Evensen, 1992; Bishop et al., 2000; Hunt et al., 2007; Houtekamer and Zhang, 2016).

In this project we borrow the Kalman filter analysis step, also known as the update step, to update the model state. Our method is not strictly speaking a Kalman filter since we do not have access to the forecast model, which means we are unable to perform the forecast step required in a Kalman filter method. This process of using the forecast model to evolve the DA corrections forward to the next assimilation time is called cycling, and is a central part of most DA systems. We view the fact that we are unable to do this as an important limitation of this study.

To get around the lack of cycling, we instead make use of a single analysis step to update an entire 7-day forecast window at once. This requires augmenting the state vector to include components from each leadtime within the window, which brings the dimension of the state space to $\sim 2 \times 10^7$. We also augment the observation vector

to include observations from each observation time. Since we are assimilating observations from the first three days of each window, the dimension of the observation space is 219.

It is important to note that this process of simultaneously assimilating observations from different times means that the method used in this study is not a filter method, since filter methods by definition assimilate observations sequentially according to the time the observations were taken. Methods in which observations from different times are assimilated simultaneously are called smoothers. The method used here, in which a Kalman filter analysis step is used to assimilate observations from a whole window of times, is quite an unconventional smoother implementation, and more conventional methods such as 4DVar and 4DEnVar are described in Bannister (2017).

The specific method whose analysis step we borrow is called the ensemble transform Kalman filter (ETKF). The implementation we use is from the parallel data assimilation framework (PDAF), described in section 3.5. The choice of method was motivated partially by the fact that the availability of the PDAF routine makes implementation easier, and partially by the fact that the ETKF is computationally efficient compared to many other DA methods (Hunt et al., 2007). The ETKF implementation provided by PDAF is based on the LETKF method described in Hunt et al. (2007), without applying localisation. Asch et al. (2016) gives a thorough description of the method, which is outlined here.

In the ETKF method, the analysis is performed in ensemble subspace rather than in the state space. Algebraically this means expressing the increment as a linear combination of ensemble perturbations

$$\mathbf{X}_f = \frac{1}{\sqrt{N-1}} \left(\mathbf{x}_1^f - \bar{\mathbf{x}}^f, \mathbf{x}_2^f - \bar{\mathbf{x}}^f, \dots, \mathbf{x}_N^f - \bar{\mathbf{x}}^f \right), \quad (2.17)$$

$$\mathbf{x}^a = \bar{\mathbf{x}}^f + \mathbf{X}_f \mathbf{w}^a, \quad (2.18)$$

where $\mathbf{x}_i^f$ is the state of ensemble member i prior to assimilation and the ensemble mean $\bar{\mathbf{x}}^f$ is used as the background state. The columns of $\mathbf{X}^f$ are the perturbations of each ensemble member from the ensemble mean, and $\mathbf{w}^a$ is a coefficient vector which determines the weighting of each ensemble perturbation in equation 2.18. From Asch et al. (2016) pp.163-164, the optimal coefficient vector $\mathbf{w}^a$ is given by

$$\mathbf{w}^a = \left(\mathbf{I}_N + \mathbf{X}_f^\top \mathbf{H}^\top \mathbf{R}^{-1} \mathbf{H} \mathbf{X}_f \right)^{-1} \mathbf{X}_f^\top \mathbf{H}^\top \mathbf{R}^{-1} \mathbf{d}_b^o, \quad (2.19)$$

where $\mathbf{d}_b^o = \mathbf{y} - \mathbf{H}\bar{\mathbf{x}}^f$. This method is algebraically equivalent to performing the analysis step in state space (equation 2.16) with background state $\bar{\mathbf{x}}^f$ and background error covariance matrix given by the forecast error covariance matrix

$$\mathbf{P}_f = \mathbf{X}_f \mathbf{X}_f^\top. \quad (2.20)$$

This matrix is rank deficient since its column space has dimension equal to the ensemble size, which limits the degrees of freedom of the increments as discussed earlier. Indeed, this limitation can also be seen directly from equation 2.18, since the vector $\mathbf{w}^a$ is in the ensemble space which has dimension equal to the number of ensemble members, much less than the dimension of the state space. This method was chosen nonetheless since according to Hunt et al. (2007), working in ensemble space has considerable advantages over working in state space in terms of computational cost.

Note that this method only gives a single analysis state, where we started with an ensemble. To get a full analysis ensemble, we first define the analysis error covariance matrix, which is prescribed by Kalman theory (Asch et al., 2016, section 3.4.3) to be

$$\mathbf{P}^a = (\mathbf{I}_n - \mathbf{K}\mathbf{H}) \mathbf{P}^f. \quad (2.21)$$

Here n is the dimension of state space as before. In this case the Kalman gain matrix takes the form $\mathbf{K} = \mathbf{P}^f \mathbf{H}^\top (\mathbf{H} \mathbf{P}^f \mathbf{H}^\top + \mathbf{R})^{-1}$. To generate the analysis ensemble, we need to factorise $\mathbf{P}^a$ to get the analysis ensemble perturbations $\mathbf{P}^a = \mathbf{X}_a \mathbf{X}_a^\top$. We can rewrite equation 2.21 as

$$\begin{aligned} \mathbf{P}^a &= \left(\mathbf{I}_n - \mathbf{X}_f \mathbf{X}_f^\top \mathbf{H}^\top (\mathbf{H} \mathbf{P}^f \mathbf{H}^\top + \mathbf{R})^{-1} \mathbf{H} \right) \mathbf{X}_f \mathbf{X}_f^\top \\ &= \mathbf{X}_f \left(\mathbf{I}_N - \mathbf{X}_f^\top \mathbf{H}^\top (\mathbf{H} \mathbf{P}^f \mathbf{H}^\top + \mathbf{R})^{-1} \mathbf{H} \mathbf{X}_f \right) \mathbf{X}_f^\top, \end{aligned}$$

so we define $\mathbf{X}_a$ as

$$\mathbf{X}_a = \mathbf{X}_f \left(\mathbf{I}_N - \mathbf{X}_f^\top \mathbf{H}^\top (\mathbf{H} \mathbf{P}^f \mathbf{H}^\top + \mathbf{R})^{-1} \mathbf{H} \mathbf{X}_f \right)^{\frac{1}{2}}. \quad (2.22)$$

The notation $\mathbf{A}^{\frac{1}{2}}$ here refers to the matrix square root, which is defined to be a matrix $\mathbf{B}$ such that $\mathbf{B}^\top \mathbf{B} = \mathbf{A}$. This definition is ambiguous since for any orthogonal matrix $\mathbf{U}$, $\mathbf{U}\mathbf{B}$ is also a square root of $\mathbf{A}$. The method described in Hunt et al. (2007) takes the symmetric square root, which is uniquely defined by Horn and Johnson (2013), theorem 7.2.6. A desirable property of this choice is that it leads to an unbiased filter, that is, the mean of the analysis ensemble is equal to the mean of the forecast ensemble (Livings et al., 2008).

The matrix $\mathbf{X}_a$ can be expressed in a simpler form by rearranging equation 2.22 to

$$\mathbf{X}_a = \mathbf{X}_f \left(\mathbf{I}_N + \mathbf{X}_f^\top \mathbf{H}^\top \mathbf{R}^{-1} \mathbf{H} \mathbf{X}_f \right)^{-\frac{1}{2}},$$

using the Sherman-Morrison-Woodbury identity (Golub and van Loan, 1996, p. 50). The full analysis ensemble can then be generated from the analysis state derived in equation 2.19 via

$$\mathbf{x}_i^a = \mathbf{x}^a + \sqrt{N-1} \mathbf{X}_f \left[\left(\mathbf{I}_N + \mathbf{X}_f^\top \mathbf{H}^\top \mathbf{R}^{-1} \mathbf{H} \mathbf{X}_f \right)^{-\frac{1}{2}} \right]_i, \quad (2.23)$$

where the notation $[_]_i$ refers to the i^{th} column of the matrix being taken. Now the analysis ensemble perturbations match exactly the form of the forecast ensemble perturbations,

$$\mathbf{X}_a = \frac{1}{\sqrt{N-1}} (\mathbf{x}_1^a - \mathbf{x}^a, \mathbf{x}_2^a - \mathbf{x}^a, \dots, \mathbf{x}_N^a - \mathbf{x}^a), \quad (2.24)$$

since the i^{th} column of $\mathbf{X}_a$ is

$$\begin{aligned} [\mathbf{X}_a]_i &= \mathbf{X}_f \left[\left(\mathbf{I}_N + \mathbf{X}_f^\top \mathbf{H}^\top \mathbf{R}^{-1} \mathbf{H} \mathbf{X}_f \right)^{-\frac{1}{2}} \mathbf{U} \right]_i \\ &= \frac{1}{\sqrt{N-1}} \mathbf{x}^a - \mathbf{x}_i^a. \end{aligned}$$

2.2.3 Practical Aspects

In this section, we discuss some of the pitfalls that can arise when implementing the ETKF, along with potential solutions. It is important to note that none of the solutions discussed here are implemented in this study, for the sake of computational simplicity and for other reasons discussed later in the section. Description of them is included nonetheless as knowledge of the potential problems helps to explain the results of this study, and the solutions are useful to inform future work.

When calculating $\mathbf{P}^f$ from the ensemble, spurious covariances may arise at long distances (Hunt et al., 2007), especially when the ensemble size is small. This is problematic as it can lead to large increments far from observation locations, which is physically unrealistic. One way to fix this is called localisation, and involves multiplying each element of the forecast error covariance matrix $\mathbf{P}_{ij}^f$ by a factor which depends on the distance between gridpoints i and j . Representing this distance with $d(i, j)$, we define the matrix $\mathbf{D}$ by $\mathbf{D}_{ij} = f(d(i, j))$ for some function $f : \mathbb{R}^+ \rightarrow [0, 1]$ of our choosing. We then define the new error covariance matrix $\mathbf{P}^{f'}$ via the following formula:

$$\mathbf{P}^{f'} = \mathbf{P}^f \circ \mathbf{D}. \quad (2.25)$$

Here $\circ$ denotes the Schur product of the two matrices, that is $\mathbf{P}_{ij}^{f'} = \mathbf{P}_{ij}^f \mathbf{D}_{ij}$. So the localisation effectively scales the covariance values by a factor between 0 and 1 determined by the function f , the aim being to suppress the spurious long-distance covariances.

The choice of f is important as it determines the nature of the localisation. In particular, f should be close to 1 when $d(i, j)$ is small and close to 0 when $d(i, j)$ is large, to suppress covariances at long distances while allowing them to exist at short distances. Hunt et al. (2007) gives a detailed description of the local ensemble transform Kalman filter (LETKF) obtained by applying localisation to the ETKF. In that study, f is chosen by setting a cutoff radius r and defining f by

$$f(d) = \begin{cases} 1, & d < r, \\ 0, & r \leq d. \end{cases}$$

It also possible to make f a continuous function such that given in Gaspari and Cohn (1999):

$$f(d) = \begin{cases} -\frac{1}{4} \left(\frac{d}{r}\right)^5 + \frac{1}{2} \left(\frac{d}{r}\right)^4 + \frac{5}{8} \left(\frac{d}{r}\right)^3 - \frac{5}{3} \left(\frac{d}{r}\right)^2 + 1, & 0 \leq d \leq r, \\ \frac{1}{12} \left(\frac{d}{r}\right)^5 - \frac{1}{2} \left(\frac{d}{r}\right)^4 + \frac{5}{8} \left(\frac{d}{r}\right)^3 + \frac{5}{3} \left(\frac{d}{r}\right)^2 - 5 \left(\frac{d}{r}\right) + 4 - \frac{2}{3} \left(\frac{d}{r}\right)^{-1}, & r \leq d \leq 2d, \\ 0, & 2d \leq r, \end{cases}$$

where r is a cutoff radius as before. This form of f has been successfully applied to EnKF and other methods in multiple studies, such as Hamill et al. (2001); Houtekamer and Mitchell (2001); Whitaker and Hamill (2002).

Another advantage of localisation in the context of the ETKF is that it goes some way to solve the problem of rank deficiency. The matrix $\mathbf{D}$ is full rank so the forecast error covariance matrix after localisation $\mathbf{P}^{f'} = \mathbf{P}^f \circ \mathbf{D}$ is full rank also, as opposed to $\mathbf{P}^f$ which is rank deficient. Thus localisation increases the degrees of freedom in the DA increments, which has the potential to improve DA performance at the cost of increased computational expenditure.

No localisation is implemented in this study. This is partly to maintain computational simplicity as much as possible, and partly because in the examination of the forecast error covariances in section 4.1 we find some long-distance covariances that do not appear to be spurious. Excluding these potentially physically realistic long-distance covariances may have a detrimental impact on DA performance, so we go with the simpler option and do not include any localisation.

Another problem that may be encountered is $\mathbf{P}^f$ being an underestimate of the true background error covariance. This is a result of sampling error that affects the estimation of $\mathbf{P}^f$ from the ensemble (Sacher and Bartello, 2008). This alone is not necessarily

undesirable, since it will lead to the analysis covariance $\mathbf{P}^a$ being smaller which reduces spread in the analysis ensemble. Ensemble spread is a commonly used metric of forecast uncertainty, so some reduction in spread is desirable.

It is possible for $\mathbf{P}^f$ to be too small, however, as by equation 2.15 a small $\mathbf{P}^f$ leads to increments which are erroneously small, and tends to compound over multiple analysis steps since a tighter analysis ensemble leads to a smaller forecast error covariance matrix at the next analysis step, which leads to a problem called filter divergence (Jazwinski (1970), pp.301-305). Note that in our case it will not compound since we are only doing one analysis step per model run, but it should be checked that the increments are realistic and the analysis ensemble spread is not too small even after one analysis step.

This can be tested by comparing the analysis ensemble to the true value. If the truth lies outside the range given by the ensemble spread, then the ensemble spread is too small and we say the ensemble is ‘overconfident’. A schematic is shown in figure 2.3. The ensemble on the left has appropriate spread, as the ensemble mean lies within the range given by the ensemble. The ensemble on the right is overconfident, as the range of the ensemble is too small to capture the truth.

Note that we do expect the truth to lie outside this range sometimes, as we expect the ensemble values to be randomly distributed around the true value. However, a high prevalence of overconfident ensembles is undesirable, since as mentioned before, ensemble spread is commonly used to estimate uncertainty in a forecast. Overconfident ensembles may lead to too much confidence being placed in an incorrect forecast, which is detrimental to any decision making that is informed by the forecast.

Various methods are available to counteract this problem, but one of the simplest is called inflation (Anderson and Anderson, 1999). This involves multiplying the forecast error covariance matrix $\mathbf{P}^f$ by a predefined factor $\alpha > 1$ prior to the analysis step. This increases the size of the increments that may be applied by the system, and also increases the spread of the analysis ensemble, which reduces overconfidence.

As with localisation, we do not apply any inflation in this study. This choice was informed by preliminary DA experiments with no inflation in which the relation between forecast error and ensemble spread seemed appropriate.

2.3 Chapter Summary

This chapter is intended to give an overview of the theoretical aspects of this study. We have seen details of ocean wave theory, including qualitative descriptions of physical mechanisms affecting waves and quantitative treatments of the linear theory and wave spectra used in WAVEWATCH III. A description of DA theory has been given, from the basic aspects of state vectors and cost functions to a more specific description of Kalman filters and some potential problems that can arise when using these methods.

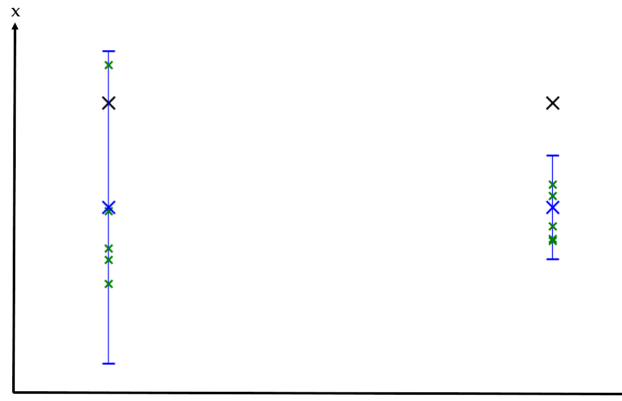


FIGURE 2.3: Left: an ensemble with appropriate spread. Right: an over-confident ensemble. Truth is shown in black, ensemble mean and spread is shown in blue and ensemble members are shown in green.

Particular attention has been given throughout to the topics most relevant to this study, such as the ETKF method which is borrowed heavily from. It is hoped that the theory presented in this chapter provides a basis for understanding the methods and results presented in the subsequent chapters of this study.

Chapter 3

Experimental Design

This chapter is intended to describe the details of the experimental design used in this project. It contains a description of the wave model used to generate the forecasts, as well as an overview of the background state and the observations available to us. Finally a description is given of the twin experiment method used in this project, as well as the data assimilation setup.

3.1 Model Description and Domain

The forecast data used in this project comes from Fugro, and this section contains a description of the process used to generate this data. The setup of the WAVEWATCH III (WW3) model used by Fugro is discussed, as well as other aspects such as the model domain and frequency of forecasts.

Waves are modelled by Fugro using WW3 (WW3DG, 2019), which uses as its base spectrum a wavenumber-direction spectrum $F(k, \theta)$, rather than the frequency-direction spectrum described in section 2.1.3. The spectrum is discretised using 29 sampled wavenumbers k and 24 angles θ . The wavenumber-direction spectrum is related to the frequency-direction spectrum by the Jacobian transform

$$F(\sigma, \theta) = \frac{\partial k}{\partial \sigma} F(k, \theta) = \frac{1}{c_g} F(k, \theta)$$

Wave propagation is then described by the equation

$$\frac{\partial}{\partial t} N(k, \theta) + (\mathbf{c}_g + \mathbf{U}) \cdot \nabla N(k, \theta) = \frac{S}{\sigma} \quad (3.1)$$

Where $N(k, \theta) = F(k, \theta)/\sigma$ as in equation 2.12 and S describes the net effect of sources and sinks for the spectrum $F(k, \theta)$. The symbol ∇ refers to the horizontal gradient i.e. $\nabla N = \frac{\partial N}{\partial x} + \frac{\partial N}{\partial y}$ with x and y horizontal Cartesian coordinates.

In deep water, S consists of three main parts: an atmospheric interaction term S_{in} , a nonlinear wave-wave interaction term S_{nl} and a dissipation term S_{ds} . S_{in} is dominated by an exponential wind-wave growth term, so S_{in} is equal to some coefficient multiplied by $\sigma N(k, \theta)$. This leads to a feedback which is usually positive but can be negative in the case of swell and represents the main source of wave energy. S_{ds} is generally dominated by wave breaking, and it represents a sink of wave energy. Similar to S_{in} , it is equal to a negative coefficient multiplied by $\sigma N(k, \theta)$. The balance between S_{in} and S_{ds} determines the overall growth or decay of wave energy.

S_{nl} represents interactions between waves due to nonlinear processes, which results in energy transfer between waves of different frequencies. The exact mechanisms behind this are poorly understood, with many theories proposed, and a general discussion of the topic is given in Janssen (2004), chapter 4. Reflecting this, WW3 gives many options for how the nonlinear source term should be calculated. The option used by Fugro is based on the generalised multiple discrete interaction approximation from Tolman (2013).

In shallow water, processes such as wave-bottom interactions, wave breaking and reflection off shorelines become important, so additional source terms are included to describe this.

From the forecasted spectra, other variables such as wave height, period and mean direction are derived. The three variables focused on in this study are significant wave height H_s , and two measures of the wave period T_{m01} and T_{m02} . These are related to the spectrum via the spectral moments (equations 2.8, 2.11).

The reason for assimilating both T_{m01} and T_{m02} is that they are defined from different spectral moments, so assimilating both will provide more information about the spectrum than either one alone, which is expected to lead to better DA results.

As forcing, WW3 uses just the 10 metre wind, and this comes from the National Oceanic and Atmospheric Association's (NOAA) global ensemble forecasting system (GEFS) (NOAA, 2023). A 31 member ensemble is taken from GEFS, and the u and v wind fields at 10m are used to force WW3. Each GEFS ensemble member provides forcing for one of the 31 WW3 ensemble members.

Fugro use a nested domain to produce the forecasts, running WW3 on a 0.5° resolution global rectangular grid, and two higher resolution nested grids covering the Gulf of Mexico. There is a 0.1° resolution grid spanning the entire Gulf, from $15.0 - 33.0^\circ\text{N}$ and $79.1 - 100^\circ\text{W}$, and a 0.05° grid spanning from $18 - 22.5^\circ\text{N}$ and $90.5 - 96.5^\circ\text{W}$. These domains are shown on a map in figure 3.1.

Forecasts are produced at 12Z each day, and run up to seven days ahead. Fugro have provided us with pre-computed forecast data for seven of these daily runs, with start dates spanning from 06/03/2022 to 12/03/2022. A schematic of this process is shown

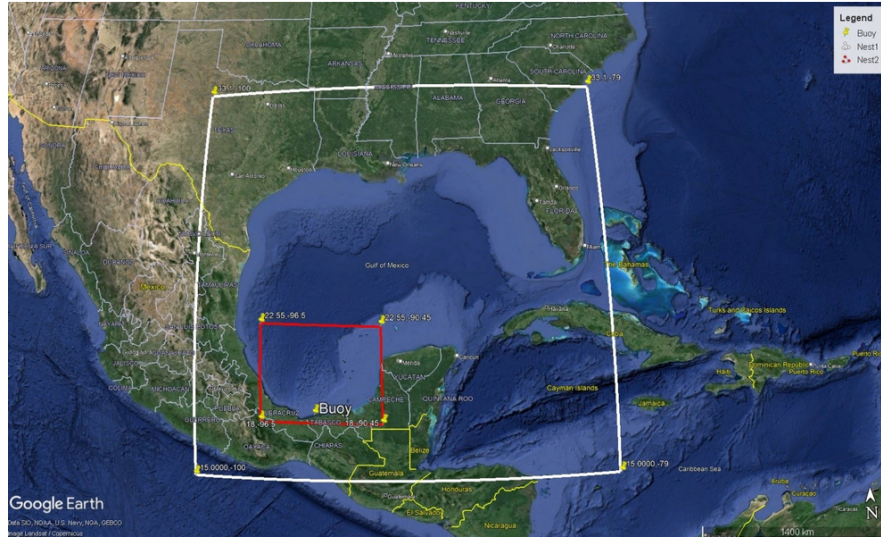


FIGURE 3.1: Model domains shown on a map. The white rectangle shows the outer domain, the red rectangle shows the inner domain and the buoy is marked with a yellow pin.

in figure 3.2. As stated in section 2.2.2, we are unable to run the forecast model in this project, so each 7-day forecast window is treated separately with a single analysis step.

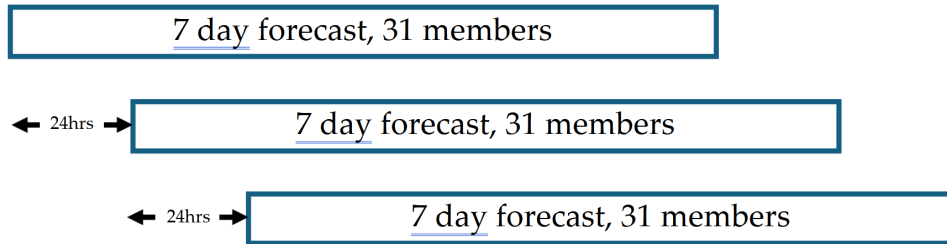


FIGURE 3.2: Schematic of the forecasting setup used by Fugro. A new 7-day 31-member ensemble forecast is produced at 12Z each day.

3.2 Background State

The background state refers to the state of the system before DA, which in our case is given by the pre-computed forecast data from Fugro. This section contains a general overview of the background state in the domain, across the period from 06/03/2022 to 19/03/2022. We identify the broad features of the waves in this window, and also discuss the effect of a cold front that passed through the region on 12/03/2022. This cold front is particularly important to consider as these events are a very common cause of high waves in the region, and in our examination of the results of this study we are particularly interested in forecasts of the waves associated with the cold front.

Figure 3.3 shows the significant wave height and direction averaged over the first six days and last six days of the forecast period. H_s is mostly between 1m and 2m in the central gulf, with lower significant wave heights of < 1 m in sheltered coastal areas such

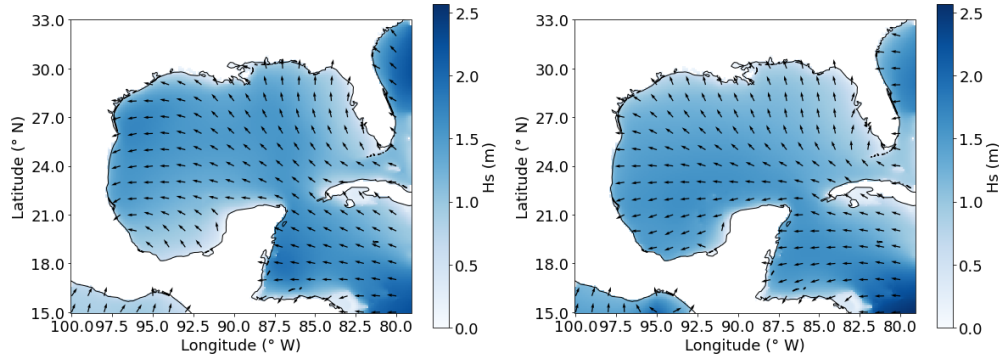


FIGURE 3.3: Left: Significant wave height (colours) and wave direction (arrows) averaged between 12Z 06/03/2022 and 12Z 12/03/2022. Right: The same averaged between 12Z 13/03/2022 and 12Z 19/03/2022

as the western side of Florida and the Yucatan peninsula. This is normal for the region according to Appendini et al. (2014), in which a 30-year wave hindcast was constructed for the GoM using the MIKE 21 SW wave model (Sørensen et al., 2004) forced by winds from the North American regional reanalysis (Mesinger et al., 2006). In that study, it was found that mean H_s in the GoM is generally in the range 1m - 1.5m, with lower values < 1m in the shallow coastal waters.

The central part of the domain has largely east-southeasterly waves over both periods, which diverge after they pass between Yucatan and Cuba. The two periods are separated by the passage of strong cold front (figure 3.4) and there is a slight shift to more easterly waves after the front has passed. This change is likely related to the wind backing associated with the front. The southern part of the Gulf of Mexico has slightly higher waves, around 1 - 1.5m, in the six day period after the passage of the front.

There is a large peak in significant wave height on the 12/03/2022, related to a cold front passing through the region. A map of H_s in the GoM at 18:00 on 12/03/2022 is shown in figure 3.4, along with a surface chart from the same time. The cold front is visible on the chart, and high waves are present in a large region behind the front. This is again in line with Appendini et al. (2014), in which cold fronts are identified as a common driver of high waves between October and April, with the most intense peaks occurring between December and March.

The height of the waves during this event is up to 6m. This is not unusual according to Panchang et al. (2013), in which wave height extremes in the GoM are analysed. The authors discuss what they call the SWH100, being the maximum estimated significant wave height in a 100 year period. They estimate SWH100 using multiple methods and in each case find that SWH100 is >10m in much of the central Gulf. The proximity of large H_s values to the Southern coast of the GoM is slightly more unusual to see, as Panchang et al. (2013) finds SWH100 to be ~ 7 m much of the Southern GoM. The H_s values seen here are still below this, but not by as much, indicating that the magnitude

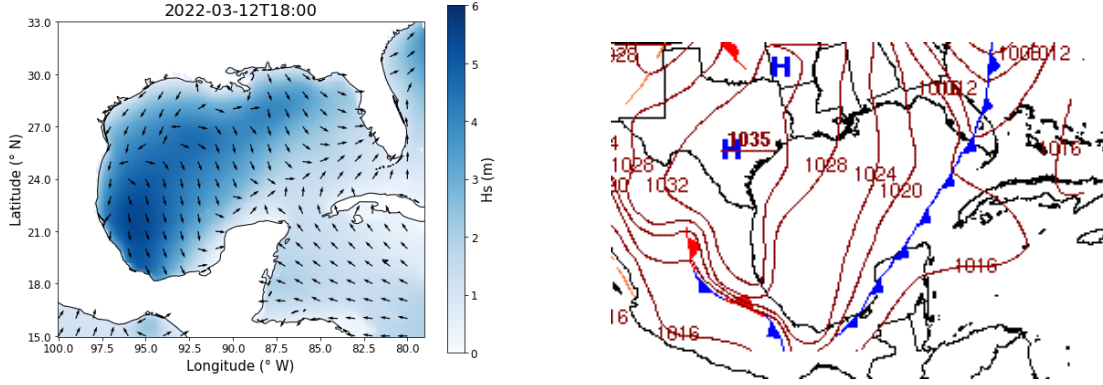


FIGURE 3.4: Left: Significant wave height (colours) and direction (arrows) in the GoM at 18Z on 12/03/2022. Right: Surface analysis chart valid for 18Z on 12/03/2022, taken from NOAA weather prediction centre (WPC) surface analysis archive (NOAA, 2022).

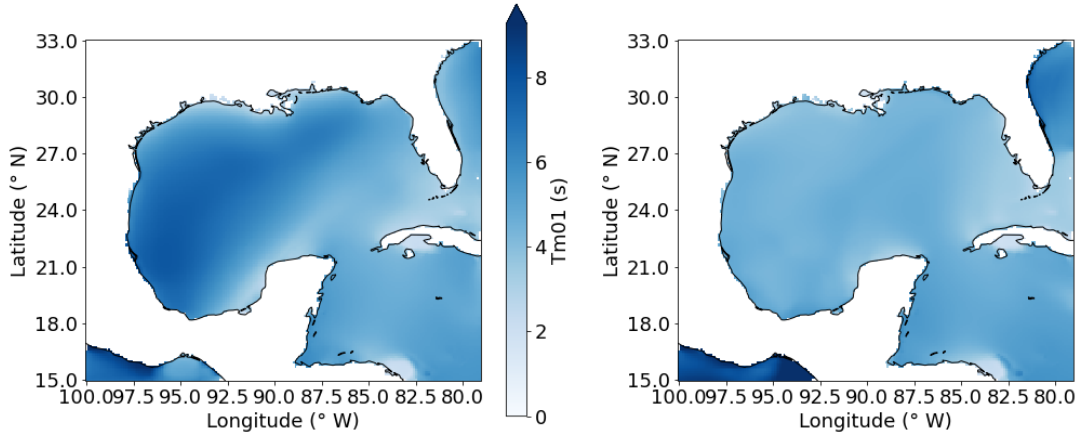


FIGURE 3.5: Left: T_{m01} at 18Z on 12/03/2022. Right: T_{m01} averaged over the period from 12Z 10/03/2022 to 12Z 12/03/2022

of significant wave heights forecast in this region may be unusual from a climatological standpoint.

Figure 3.5 shows T_{m01} at 18Z 12/03/2022 and averaged over the preceding 48 hours. The high waves associated with the cold front at 18Z have a period of ~ 8 s, which is higher than the period in the rest of the GoM at that time and also higher than the waves across the entire GoM in the 48 hours prior. This is also high compared to climatology, since Appendini et al. (2014) finds the average period in the region to be ~ 6 m.

In summary, we have seen that the significant wave heights during the forecast window were normal for the region, with the exception of those induced by a cold front on 12/03/2022. The waves associated with this cold front had longer period than most waves in the region, and the significant wave heights seen in the Southern GoM were somewhat climatologically unusual. We view this cold front event as particularly worthy of analysis, since it is this kind of event that poses a danger to offshore operations.

This is especially true looking forward since Appendini et al. (2014) determined that extreme wave heights in the region are becoming higher over time.

3.3 Buoy Observations

This section gives an overview of the buoy observations used in this study. The observations are a crucial part of DA as they give information about the true state of the system which can be used to improve forecasts. The equipments and methods used to generate the observations are discussed, as well as patterns and properties of the observational data.

Our observations come from an acoustic Doppler current profiler (ADCP) buoy (Boyd, 2006), which estimates the spectrum by measuring the wave induced surface height variations and velocity at different depths. According to Vallis (2017), the velocity and surface height anomaly induced by a simple wave with wave vector $\mathbf{k} = (k_x, k_y)$, frequency σ and amplitude a is

$$\begin{aligned} u &= a \frac{k_x}{\sigma} g \exp i(\mathbf{k} \cdot \mathbf{x} - \sigma t) \cosh k(z + D), \\ v &= a \frac{k_y}{\sigma} g \exp i(\mathbf{k} \cdot \mathbf{x} - \sigma t) \cosh k(z + D), \\ w &= -ia \frac{k}{\sigma} g \exp i(\mathbf{k} \cdot \mathbf{x} - \sigma t) \sinh k(z + D), \\ \eta &= a \exp i(\mathbf{k} \cdot \mathbf{x} - \sigma t), \end{aligned}$$

where only the real part of each expression should be taken. Note that the horizontal velocity (u, v) is always in the same direction as the wave vector $\mathbf{k}$. From the ADCP measurements, the cross-covariance between the velocities measured at different depths may be combined to give an estimate of the direction-integrated spectrum $S(\sigma) = \int F(\sigma, \theta) d\theta$. Further details are given in Boyd (2006), chapter V, in which the direction-integrated spectrum is notated $E(\omega)$. Bulk wave variables such as those assimilated in this study can then be derived from the spectral moments as described in section 2.1.3

Observations are available over a roughly 6 month period spanning from October 2021 to April 2022, which includes the period for which forecasts are available. A timeseries of the observed significant wave height from the forecast period is included in figure 3.6. The peak in significant wave height late on the 12th of March is clearly visible on the plot.

Figure 3.7 shows a timeseries of observed H_s and T_{m01} from a longer period. It shows that outside of specific peaks, H_s is generally $< 1\text{m}$. This is consistent with the model background state which showed H_s in the buoy region to be $< 1\text{m}$ for much of the forecast period. The peaks generally consist of a sharp increase in H_s followed by a

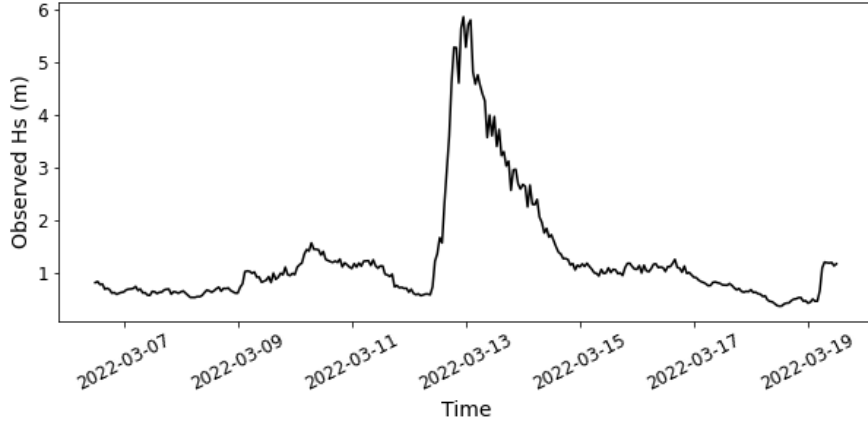


FIGURE 3.6: Observed timeseries of significant wave height from ADCP buoy

more gradual decrease down to the baseline level. This is consistent with the slightly higher waves seen in the model background state in the buoy region after the passage of the front. As seen in the background state, the peaks in H_s are associated with peaks in T_{m01} . This points to a positive correlation between H_s and T_{m01} which is discussed further in section 4.1.

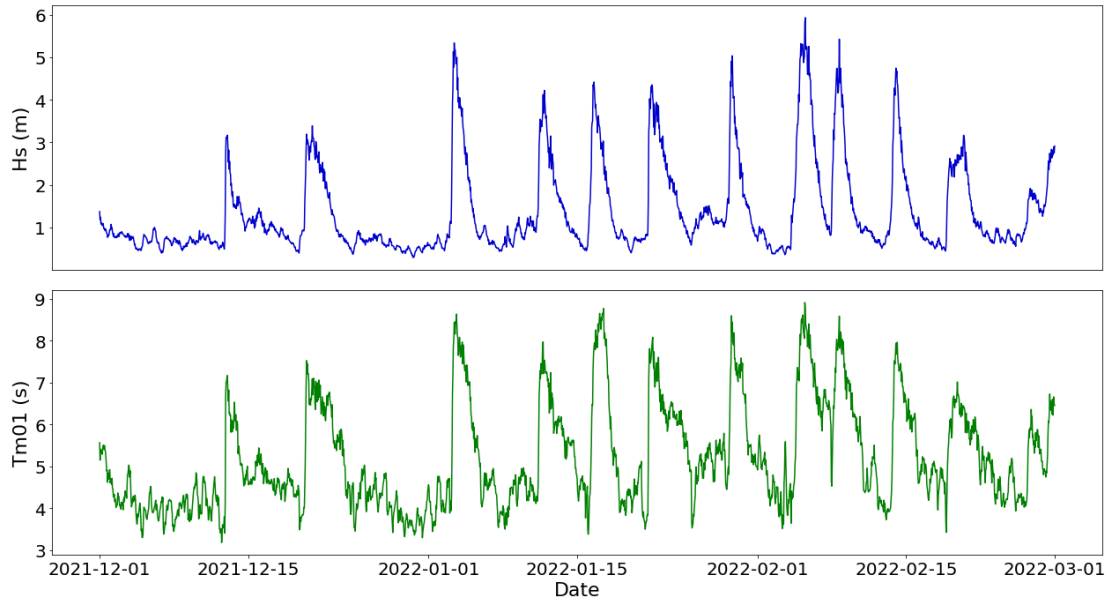


FIGURE 3.7: Timeseries of observed significant wave height (top) and T_{m01} (bottom) over a three-month period from 01/12/2021 to 01/03/2022

Figure 3.8 shows a density plot of observed H_s and wave direction at the buoy location. The colour of each region on the plot shows the number of observations in that region. There is a clear peak in frequency corresponding to waves of height $\sim 0.5\text{m}$ coming from a northeasterly direction. Some higher waves (up to $\sim 5\text{m}$) are present coming from a north-northwesterly direction. This lines up with the direction of waves produced by the cold front on 12/03/2022, suggesting that the waves produced by this

kind of event consistently reach the buoy from a north-northwesterly direction. Since the buoy is near the coast in the southern part of the gulf, it is sheltered from the south so there are virtually no waves coming from this direction.

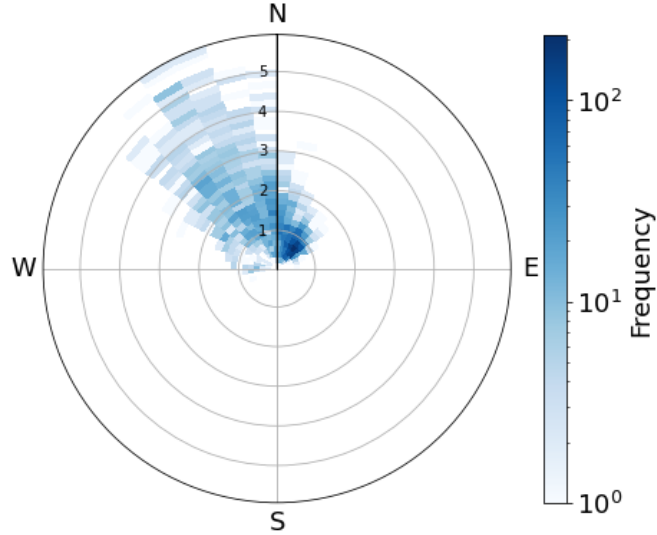


FIGURE 3.8: Density plot of significant wave height (radial axis) and wave direction at the buoy location.

3.4 Twin Experiment

A twin experiment, also known as an observing system simulation experiment (OSSE), is a popular methodology often used for evaluating the impact of new observing systems when observational data is not yet available (Zeng et al., 2020). Even though we do have observations available, the OSSE methodology is used because it makes the ‘truth’ available across the entire domain, which is not the case when assimilating real observations. This is desirable because it allows forecast error to be calculated away from the positions and times of observations.

In an OSSE, data from a model run (often called the ‘nature run’, here called the ‘truth’) is assimilated in place of real observational data. Here we select one of the 31 ensemble members to represent the truth, and DA is applied to the remaining 30 ensemble members. The same ensemble member is selected for each forecast window, to maximise continuity in the generated observations. Since the forecast windows overlap, the ‘truth’ contains multiple values for each time depending on the forecast window, and these values may differ slightly. This is not a problem for this study, since each forecast window is treated as its own DA problem with its own ‘truth’ (see section 3.5 for further discussion).

The observations are generated by sampling values from the truth at predefined observation locations and times. Hourly values are sampled from the nearest gridpoint to the buoy to match the real observations that are available. To represent observation error, some random noise is added to the sampled values. This noise is drawn from

a normal distribution with standard deviation equal to the error standard deviation derived in section 4.2

A limitation of the OSSE method is its inability to correct biases in the model. Figure 4.10 shows a possible bias in the forecasted values of H_s , which would be desirable to try to correct with DA. Since the data assimilated in this project is also from model output, it will be subject to all the same biases as the forecast data so when generating the observations as described in the previous paragraph it is not possible for the DA to correct these biases. It is possible to add an artificial bias when generating the observations, but this is not pursued in this project in the interests of simplicity.

3.5 Data Assimilation Setup

To perform the data assimilation, we use pyPDAF (Chen et al., 2024), which is a python interface for the Fortran-based Parallel Data Assimilation Framework (PDAF). We will use the analysis step from the ETKF method which comes builtin with PDAF. As previously stated, we do not have access to WW3, which makes a conventional forecast step impossible.

To get around this, at each analysis step an entire 7-day forecast window is corrected, and in place of the forecast step data is loaded from the next forecast window. A schematic of the process is included in figure 3.9. So, in our setup, each 7-day forecast window is treated as a stand-alone DA problem with its own truth. One ramification of this is that improvements in one window are not propagated to the next.

Since observations are assimilated from multiple times, this requires the use of 4D state vectors as mentioned in section 2.2.1. This means the state vector contains components from each timestep within a forecast window i.e.

$$\mathbf{x} = \begin{pmatrix} \mathbf{x}(t = 0) \\ \mathbf{x}(t = 1 \text{ hrs}) \\ \mathbf{x}(t = 2 \text{ hrs}) \\ \vdots \\ \mathbf{x}(t = 7 \text{ days}) \end{pmatrix}.$$

In this way, leadtime within a forecast window is treated the same as the two spatial coordinates by the DA system. Since there are 169 timesteps in each window, and three state variables on a 181×210 grid, the dimension of the state space is approximately 2×10^7 .

At each analysis step, observations are assimilated from the first three days of the forecast window. This time period is referred to as the ‘assilimation window’. This allows the calculation of errors at times after the assimilation window and the assessment of the capability of DA to improve forecasts. This means assimilating observations of the

three state variables (H_s , T_{m01} and T_{m02}) from the first 73 timesteps of each window, so the dimension of the observation space is 219.

Since extremes of H_s are the most dangerous wave events likely to be faced in offshore operations, H_s is considered to be the most important variable out of the three assimilated. This gives motivation to perform two DA experiments: one in which H_s alone is assimilated, referred to as H_s DA, and one in which the three variables are assimilated together, called all vars DA.

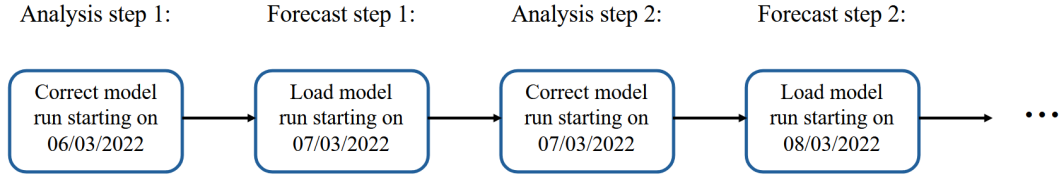


FIGURE 3.9: Schematic of the forecast and analysis steps from the data assimilation method used in this study.

3.6 Chapter Summary

In this chapter, we have seen details of the experimental design used in this study, including the methods used to generate the forecasts and observations used and the methods of the DA experiments performed. Descriptions of the background state given by the forecast data and the nature of the observations used are also given. In subsequent chapters, the scientific findings of this study will be presented and analysed, including the nature of the background and observation errors and the results of the two DA experiments.

Chapter 4

Background and Observation Errors

This chapter contains a discussion of the background and observation errors in our DA system. Though the background errors are determined directly from the ensemble when using the ETKF method, understanding the form they take is crucial to understanding how the DA system works. This is because, according to equation 2.16, the DA increments always lie in the column space of the background error covariance matrix. This means that the background errors determine how information from the observations can be propagated around the domain by DA.

The observation errors are also important to understanding a DA system since the balance between observation and background errors determines how much weight is given to the observations by the DA. The observation errors must be estimated manually, and a description of the methods used to do this are included here.

4.1 Background Errors

In our case the background error covariance matrix is approximated by the forecast error covariance matrix $\mathbf{P}^f$, defined in equation 2.20. The elements of $\mathbf{P}^f$ are the ensemble covariances at each pair of gridpoints and timesteps,

$$\mathbf{P}_{ij}^f = \text{cov} \left(\mathbf{x}_i^f, \mathbf{x}_j^f \right), \quad (4.1)$$

where $\mathbf{x}_i^f$ refers here to the ensemble values in component i of the state vector. The diagonal elements are then the ensemble variance at each gridpoint and timestep:

$$\mathbf{P}_{ii}^f = \text{var} \left(\mathbf{x}_i^f \right). \quad (4.2)$$

Figure 4.1 shows a timeseries of H_s ensemble mean and variance at the buoy location from the forecast starting 12Z 10/03/2022. A peak on the 12th is apparent as in figure 3.4. Figure 4.1 also shows that the variance is a strong function of the mean. The variance in H_s peaks just before the peak in the mean and drops back to a lower value

as the mean reaches its peak. This means that the background error in H_s is highest just before the peak in H_s .

Figure 4.2 shows the same as 4.1 but for T_{m01} . Again it shows a peak in the variance just before the peak in the mean. In this case the peak in T_{m01} is preceded by a dip, and this dip is also associated with a peak in the ensemble variance. This means that the background error variance in T_{m01} is high for a period of ~ 24 hours, as opposed to ~ 12 hours for H_s .

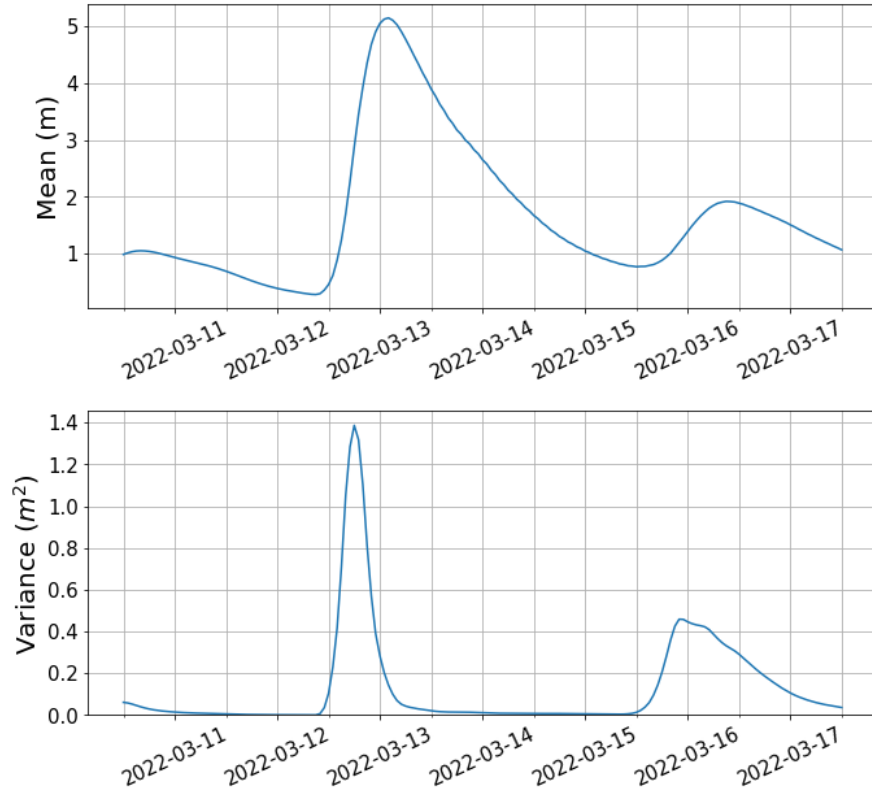


FIGURE 4.1: Timeseries of forecast significant wave height ensemble mean (top) and variance (bottom) at the buoy location from the model run starting on the 10/03/2022.

We now present maps of the ensemble covariances in order to illustrate the forms these covariances can take and gain insight into the workings of our DA system. Covariances in space will be considered, as well as cross-covariances between different variables and lagged in time. Arrows showing wind/wave direction will be used to explain the form the covariances take.

Figure 4.3 shows the spatial covariance of H_s at 12Z 12/03/2022. The colours on the map show the covariance between H_s at each point in the domain and H_s at the buoy (marked with a cross). The arrows show the direction and magnitude of the wind, averaged over the preceding 24 hours. The strongest covariances are confined to the western GoM and do not extend very far from the buoy. From the wind arrows, the wind in the region of strong covariance is very low, similar to the wind at the buoy, and that the wind in regions of weak covariance is high.

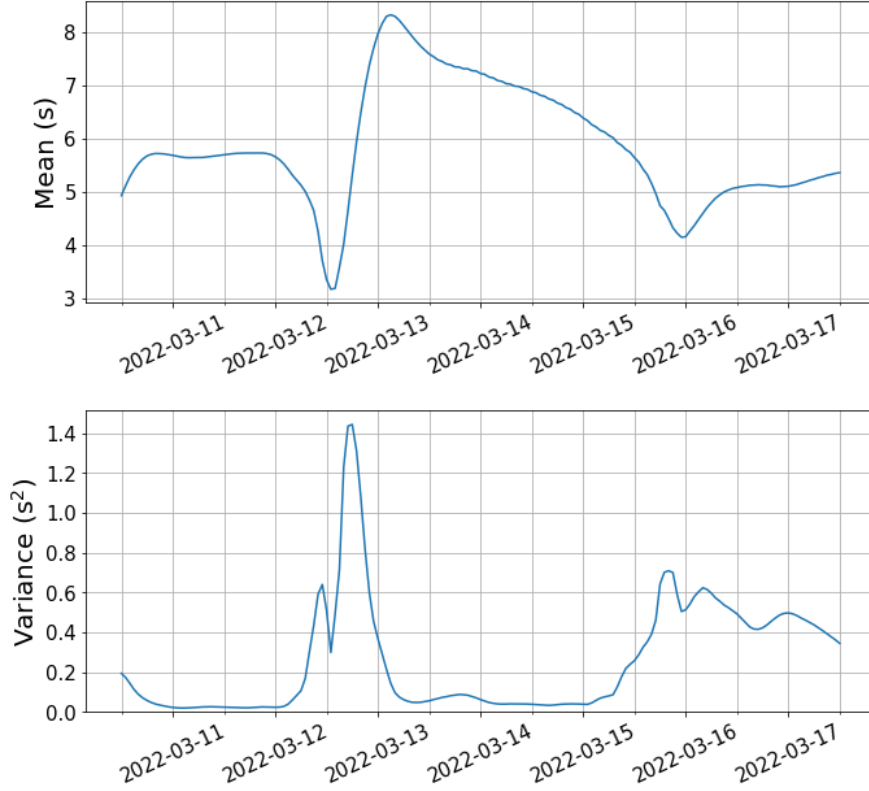


FIGURE 4.2: Timeseries of forecast T_{m01} ensemble mean (top) and variance (bottom) at the buoy location from the model run starting on the 10/03/2022.

This suggests that the wind forcing plays a role in determining the nature of the ensemble covariances. This is unsurprising since according to Lefèvre and Aouf (2012), ocean wave prediction is primarily a forcing problem, and so we expect the nature of the forcing to play a major role in determining the properties of the waves. This is the motivation for averaging the wind values over the preceding 24 hour window, since the wind forcing takes time to have an impact on the waves.

The wind forcing can also induce covariances at longer distances. Figure 4.4 shows the same as 4.3 but at 12Z 11/03/2022. The difference in magnitude of the covariance from figure 4.4 is notable. This could be explained by the fact that the variance in H_s at 12Z 11/03/2022 is much smaller than the variance at 12Z 12/03/2022, as seen in figure 4.1. In figure 4.4, nonzero covariances are visible far from the buoy in the Northern GoM. These covariances could also be present in figure 4.3, but washed out due to the larger overall magnitude of covariance in that figure. The pattern of winds gives a plausible explanation for these long-distance covariances: the region of positive covariance in the Northeast has weak winds, similar to the vicinity of the buoy, and the region of negative covariance in the Northwest has strong winds.

It is also possible that these long distance covariances are physically unrealistic. As discussed in section 2.2.3, an inherent limitation in estimating the background error from the ensemble is that a finite ensemble size leads to sampling error in $\mathbf{P}^f$. This can

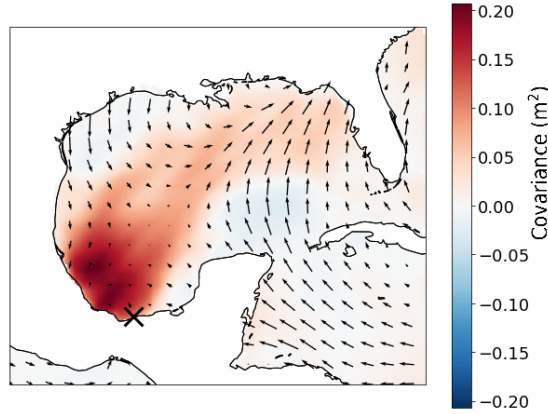


FIGURE 4.3: Colours showing spatial covariance of significant wave height at the buoy with significant wave height in the field at 12Z 12/03/2022. Arrows showing wind averaged over prior 24 hours.

cause spurious covariances to arise at long distances (Hunt et al., 2007), which can be fixed with localisation. This possibility is not explored in this study, partly due to the fact that plausible physical explanations exist for the long-distance covariances, but this is something that should be included in future work on the topic.

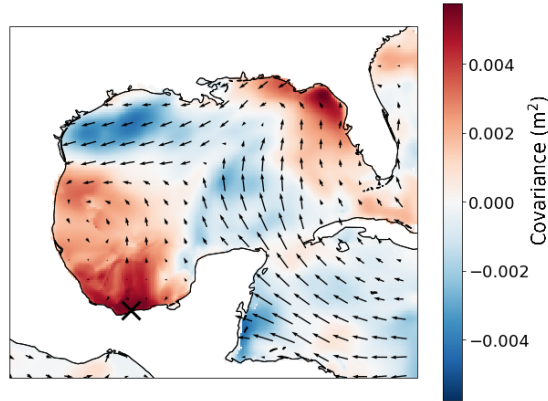


FIGURE 4.4: Colours showing spatial covariance of significant wave height at the buoy and significant wave height in the field at 12Z 11/03/2022. Arrows showing wind averaged over prior 24 hours.

Another thing that can cause long-distance covariances is travelling swell waves. Figure 4.5 shows the H_s covariance at 18Z 13/03/2022 with a one day lag. In other words, expressing H_s as a function of space and time $H_s = H_s(x, y, t)$ we plot

$$\text{cov}(H_s(x_0, y_0, t_0), H_s(x, y, t_0 - 1 \text{ day}))$$

for each x and y . Here (x_0, y_0) is the position of the buoy, and t_0 is 18Z 13/03/2022. This means the H_s ensemble at the buoy is sampled at 18Z 13/03/2022 and the H_s ensemble elsewhere in the domain is sampled at 18Z 12/03/2022, to show the correlation with earlier sea states. There is a large region of positive covariance spanning the Western GoM and much of the Northern part also. The arrows mark the wave direction at 18Z

13/03/2022, and this shows that waves from the Northwestern GoM travel West and South towards the buoy. This means they are able to influence the sea state at the buoy, which explains the positive covariance.

Another notable aspect of figure 4.5 is the general magnitude of covariance seen. The maximum covariance seen in figure 4.5 is 0.03, compared to 0.20 in figure 4.3. This is explained by the one day lag used to generate figure 4.5. We expect the covariances between two points to decrease the further apart those two points are in time, and this is what we see. This means that we expect the observations assimilated to have more of an impact on the state at times near the observation time.

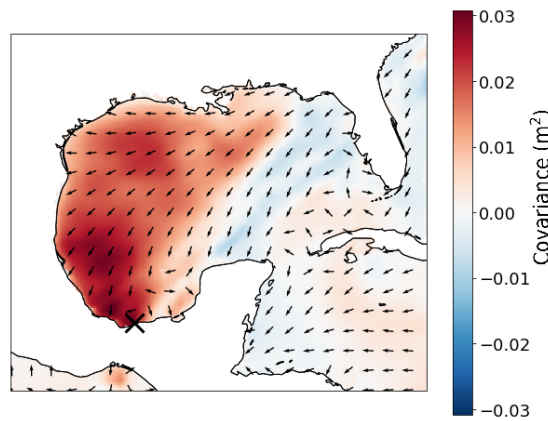


FIGURE 4.5: Colours showing covariance of significant wave height at the buoy at 18Z 13/03/2022 with significant wave height in the field at 18Z 12/03/2022. Arrows showing wave direction at 18Z 12/03/2022

So far we have only examined the covariance of H_s with itself. There are also covariances present between the different variables assimilated in this study, and this is important for the DA. These cross-covariances allow assimilation of one variable to affect other variables than the one assimilated. This will allow the assimilation of H_s alone to impact T_{m01} and T_{m02} , and allow the assimilation of all three variables to produce different forecasts of H_s than would be obtained by assimilating H_s alone. It is hoped that these cross-covariances will improve DA performance by allowing each observation assimilated to provide more information than it would in their absence.

Figure 4.6 shows the cross-covariance between H_s at the buoy and T_{m01} elsewhere at 12Z 12/03/2022. There is a large area of positive covariance to the Northwest of the buoy, with the wave direction (left plot) pointing mostly towards the buoy in this region. This suggests that the positive covariance may be related to travelling swell waves, since large swell waves with long period in the region Northwest of the buoy could travel towards the buoy and lead to higher waves there. However more investigation is required to determine if this is actually the case.

There are also small patches of strong negative covariance, directly East of the buoy and also South in the Pacific ocean. Since there is no obvious difference in the winds in each region to explain this, these negative covariances could be spurious. This is

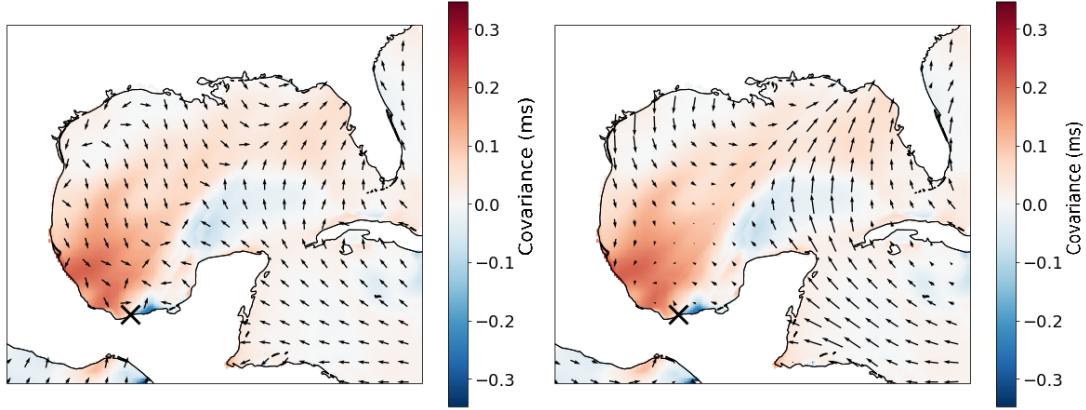


FIGURE 4.6: Cross-covariance between significant wave height at the buoy and T_{m01} in the field at 12Z 12/03/2022. Arrows show wave direction (left) and wind (right).

especially true of the negative covariance seen in the Pacific, since there is no way for the waves here to interact directly with the waves at the buoy.

In this section, we have presented maps showing the forms that the forecast error covariances can take, and attempted to explain them by referring to physical explanations from winds and wave directions, and also to sampling error. However, the physical explanations discussed here are quite simple and surface-level, and more investigation would be useful to explain the form of the covariance. Since understanding the physical mechanisms behind the covariances is not completely central to this section's goal of explaining DA performance, it is not explored further here, but it would be a good avenue for future work.

4.2 Diagnosis of Observation Errors

The observation error covariance matrix $\mathbf{R}$ is a central part of the DA system as it determines how much weight is given to each observation: if the error associated with a particular observation is large, then that observation will be given less weighting by the DA than observations with small associated errors. Unlike the forecast error covariance matrix $\mathbf{P}^f$, $\mathbf{R}$ must be estimated manually, and this section outlines the methods used to do so.

The observation error covariance matrix $\mathbf{R}$ contains the observation error covariances at each pair of observation times,

$$\mathbf{R}_{ij} = \text{cov}(\delta \mathbf{y}_i, \delta \mathbf{y}_j),$$

where $\delta \mathbf{y}_i$ represents the error in the i^{th} observation. To simplify computation, the observation error covariance matrix $\mathbf{R}$ is assumed to be diagonal, essentially ignoring

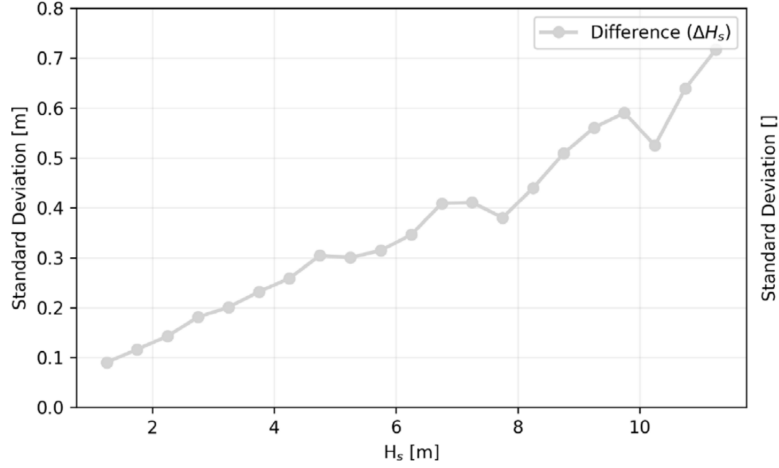


FIGURE 4.7: Observation error standard deviation in significant wave height as a function of significant wave height. Figure 2 from Houghton et al. (2023)

covariances between observations taken at different times. This also makes determining $\mathbf{R}$ easier since all that is required is to estimate the observation error variance at each point. The formula for $\mathbf{R}$ becomes

$$\mathbf{R}_{ij} = \begin{cases} \text{var}(\delta \mathbf{y}_i) & i = j, \\ 0 & \text{otherwise.} \end{cases}$$

4.2.1 Scaling Error Variance

It is not obvious how to estimate this error variance, and several methods are available for doing so. One possible method is that used in Houghton et al. (2023). They used a collocation study to estimate the error in significant wave height observations from a distributed network of drifting sofar spotter buoys. Pairs of simultaneous observations from locations within 50km of each other were collected and binned according to the average significant wave height of each pair. The difference of each pair was taken, and the standard deviation of the differences from each bin was computed. They found a strong dependence of error on significant wave height, shown in figure 4.7.

Because of this, they estimated the observation error variance not as a constant value, as in section 4.2.2, but as a constant multiple of the observations themselves - in their case 6.5%. A similar method is now applied to our observations.

In our case, the observations come from one buoy and they are available hourly, so the pairs are taken to be observations one hour apart. The error standard deviation is calculated using the method described above to obtain the plot in figure 4.8. As in Houghton et al. (2023), there is a strong linear relationship between errors and significant wave height, with the error standard deviation being approximately 9.6% of the

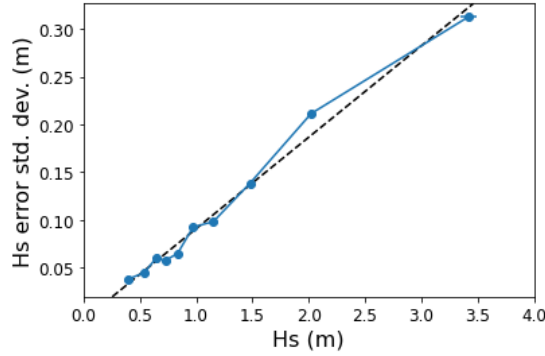


FIGURE 4.8: A plot of significant wave height error standard deviation against significant wave height. Calculated using observations from a period spanning 05/10/2021-13/04/2022

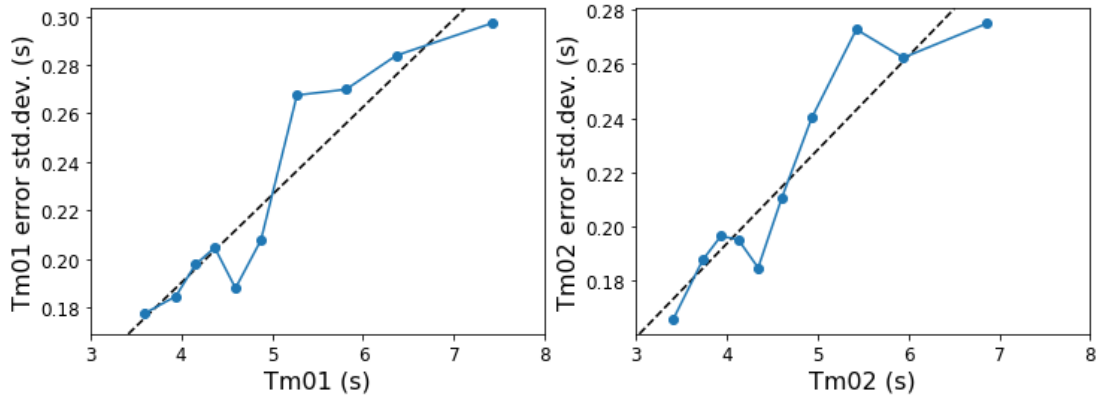


FIGURE 4.9: Plots of T_{m01} and T_{m02} error standard deviation against T_{m01} and T_{m02} respectively.

significant wave height. This value was calculated by applying a standard linear regression to the data points plotted in figure 4.8 and is larger than that in Houghton et al. (2023).

Recall from figure 4.1 that the ensemble variance H_s peaks just before the peak in H_s and drops back to lower values as H_s is at its highest value. Since ensemble variance is a good measure of the magnitude of the background error, the background error is large compared to observational error just before the peak. This means that observations from this period will be given higher weighting than most. During and shortly after the peak, the opposite is true: background error is small compared to observational error, so observations from this period will be given lower weighting.

Repeating this process with the observations of T_{m01} and T_{m02} yields the plots shown in figure 4.9. The error distributions for T_{m01} and T_{m02} also fall approximately on straight lines, though the fit is not as good as for H_s . The gradients are also shallower, with the error standard deviations being approximately 3.5% of the observed variable for both.

Since the error standard deviations match the linear model well, this motivates modelling the error standard deviation in H_s as being equal to 10% of H_s , and the error

standard deviation in T_{m01} and T_{m02} as 3.5% of T_{m01} and T_{m02} respectively. Since the errors in period are such a small percentage of the observations, it is possible that at short periods the error may become very small, which would lead the DA system fitting the observations too closely and producing physically unrealistic increments. To avoid this, we choose a minimum error threshold ε_0 and set the observation error standard deviation equal to this threshold whenever the scaling method gives a value less than ε_0 . This requires estimating a constant scale for the error variance, and methods for doing that are described in the next section.

4.2.2 Constant Error Variance

A simple way to get a constant estimate of error variance is to plot a histogram of observation - ensemble mean, called an $o - b$ plot. The spread in $o - b$ comes from a combination of observation and background error (Desroziers et al., 2005), so a simple way of estimating the observation error is to assume that observation error accounts for a constant proportion of this spread. It is reasonable to assume that the contributions from background and observation errors are similarly sized, so the observation error can be estimated to be 50% of the total spread.

An $o - b$ histogram for significant wave height is shown in figure 4.10. The spread on the histogram is approximately 1m, so the standard deviation of observational errors in significant wave height may be estimated to be approximately 50cm.

Another thing that is visible from the $o - b$ plot is a bias in either the H_s forecasts or observations. The peak of the histogram is centred at around 30cm, which means that the forecast model tends to predict significant wave heights that are about 30cm less than those observed. If this bias is a result of the model, it would be desirable to investigate the capability of DA to reduce this bias. However, with the twin experiment method that is used in this project, this would require introducing an artificial bias to the observations, which is not done in this study (see section 3.4 for details).

Desroziers et al. (2005) provides a more rigorous approach to estimating observational error variances, by defining $\mathbf{d}_b^o = \mathbf{y} - \mathbf{H}\mathbf{x}_b$ and using the relation

$$\mathbb{E} \left[\mathbf{d}_b^o (\mathbf{d}_b^o)^\top \right] = \mathbf{H}\mathbf{P}^f\mathbf{H}^\top + \mathbf{R} \quad (4.3)$$

Taking traces and dividing by the number of observations gives a good estimate for the average observation error variance $\overline{\text{var}(\delta\mathbf{y}_i)}$ which is

$$\overline{\text{var}(\delta\mathbf{y}_i)} = \frac{\text{Tr}(\mathbf{R})}{\# \text{ observations}} \approx \frac{\mathbb{E} \left[\sum_i (\mathbf{d}_b^o)_i^2 \right]}{\# \text{ observations}} - \frac{\text{Tr}(\mathbf{H}\mathbf{P}^f\mathbf{H}^\top)}{\# \text{ observations}}. \quad (4.4)$$

In our case, the observation operator is very simple and just selects the forecast values from the appropriate location and times, so the diagonal elements of $\mathbf{H}\mathbf{P}^f\mathbf{H}^\top$ are

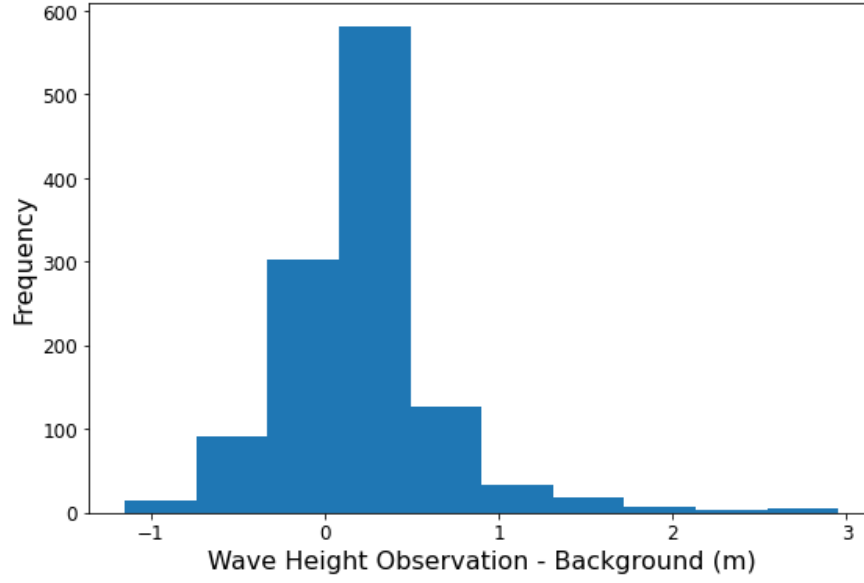


FIGURE 4.10: Observation - background histogram for significant wave height, from the forecast window 06/03/2022 to 19/03/2022.

simply the forecast ensemble variances at the observation location and times. To determine the minimum error threshold, the estimate in equation 4.4 is evaluated using only the observations of period with a value of less than 5 seconds, as these are the observations which may lead to excessively small errors. The terms on the right hand side of 4.4 are found to be

$$\frac{\mathbb{E} \left[\sum_i (\mathbf{d}_b^o)_i^2 \right]}{\# \text{ observations}} = 0.29 \text{ s}^2, \quad \frac{\text{Tr} (\mathbf{H} \mathbf{P}^f \mathbf{H}^\top)}{\# \text{ observations}} = 0.17 \text{ s}^2$$

So the observation error variance is estimated to be $0.29 \text{ s}^2 - 0.17 \text{ s}^2 = 0.12 \text{ s}^2$ and the standard deviation threshold is set at $\sqrt{0.12 \text{ s}^2} = 0.35 \text{ s}$. Examining the $o - b$ histograms for T_{m01} and T_{m02} in figure 4.11 confirms that this is a sensible value to choose.

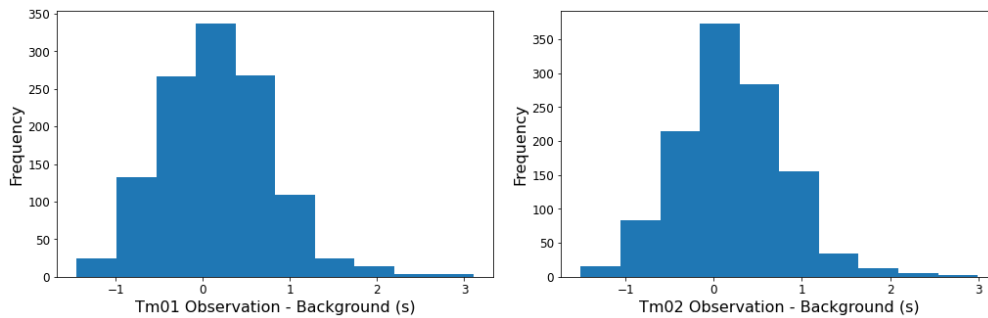


FIGURE 4.11: Observation minus background histograms for T_{m01} (left) and T_{m02} (right).

In this section, we have found that the observation error standard deviations in all variables are strong functions of the observed values, with the error in H_s being approximately 9.6% of the observed value and the error in both periods being approximately

3.5% of the observation.

Motivated by this, the H_s error is set to 10% of the observed value in the assimilation of significant wave height alone. For the assimilation of all variables, there is more risk of fitting the observations too closely as more observations are assimilated. Because of this, the scales of error standard deviation are increased from those found above to 12% for significant wave height and 7% for the periods. A minimum error threshold of 0.35s is implemented for the periods.

4.3 Chapter Summary

In this chapter we have given characterisations of the background and observation errors. Knowledge of these errors is vital to DA as between them they control a large part of the workings of a DA system. The results shown in this chapter are intended to aid understanding of the results of the DA experiments, which are presented in the next chapter.

Chapter 5

Results

In this chapter, the results of the DA experiments are presented and discussed. We have one experiment (H_s DA) where H_s alone is assimilated, and one experiment (all vars DA) where H_s , T_{m01} and T_{m02} are assimilated together. The results of these experiments are compared to one another and to the forecast data with no DA applied. We will examine properties of the results such as the increments, RMSE and ensemble spread, as well the forecasts of the H_s maximum on 12/03/2022.

Throughout this chapter, plots of H_s and T_{m01} are included. The plots of T_{m02} all look very similar to the equivalent plots of T_{m01} , so they are not required for a discussion of the results. Hence they are not included in this chapter to aid readability. For completeness, they are included in appendix A and may be referred to there.

We begin with a visual examination of timeseries to see qualitatively what effect the DA has had. Figure 5.1 shows a timeseries of H_s at the buoy from all vars DA, H_s DA, no DA and the truth. This is taken from the 7-day forecast window starting at 12Z 09/03/2022. Observations from the first three days (the ‘assimilation window’) of this forecast window are assimilated, and the end of this period is marked with a dotted blue line.

The DA has little effect during the assimilation window, since the no DA forecast lies very close to the truth here. They diverge after the end of the assimilation window (at 3 days leadtime), and H_s DA closely tracks the truth while all vars DA is closer to no DA. This is likely due to the effects of the different variables assimilated cancelling each other out. The four runs come back together at 4-5 days leadtime and all four diverge after 5 days leadtime.

These periods when the runs diverge reflect when we expect to see large forecast errors. 3-4 days leadtime contains the swell event which we have seen is associated with large forecast errors, and 5+ days leadtimes is at the end of forecast window when the error tends to be largest due to chaotic effects. That H_s DA matches the truth well at 3-4 days and not at 5+ days can be explained by the fact that 5+ days is further from the assimilation window. This means that any correction at 5+ days relies on cross-covariances in time, which are weak as seen in section 4.1.

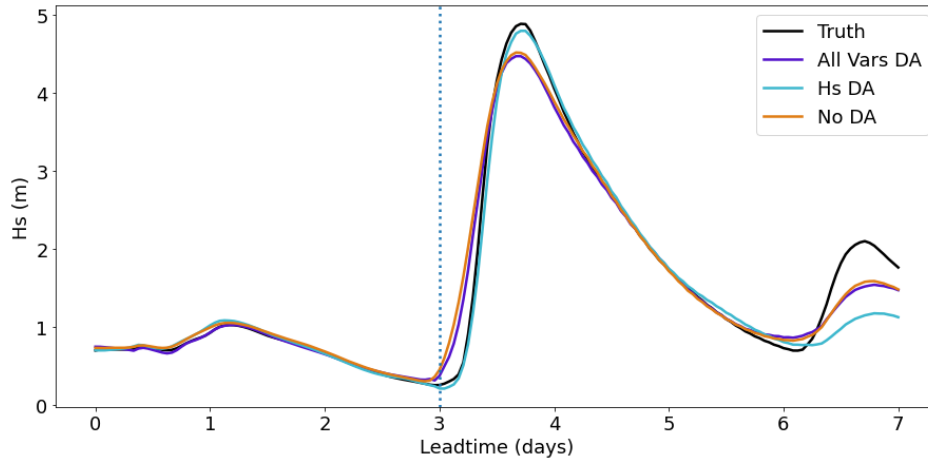


FIGURE 5.1: Timeseries of significant wave height at buoy location from 12Z 09/03/2022 model run. Truth is shown in black, and all vars DA, H_s DA and no DA ensemble means are shown in dark blue, light blue and orange respectively. Blue dotted line indicates the end of the assimilation window.

As well as the buoy location, we also examine the equivalent timeseries at a location far from the buoy. Figure 5.2 shows the same as 5.1 at $30^\circ \text{N } 80^\circ \text{W}$, which is outside the Gulf of Mexico in the Atlantic ocean near Florida. Here there is no noticeable improvement from DA, as all vars DA, H_s DA and no DA all diverge from the truth by similar amounts. This is intuitively unsurprising since we would not expect observations at the buoy to be useful in predicting the waves so far away.

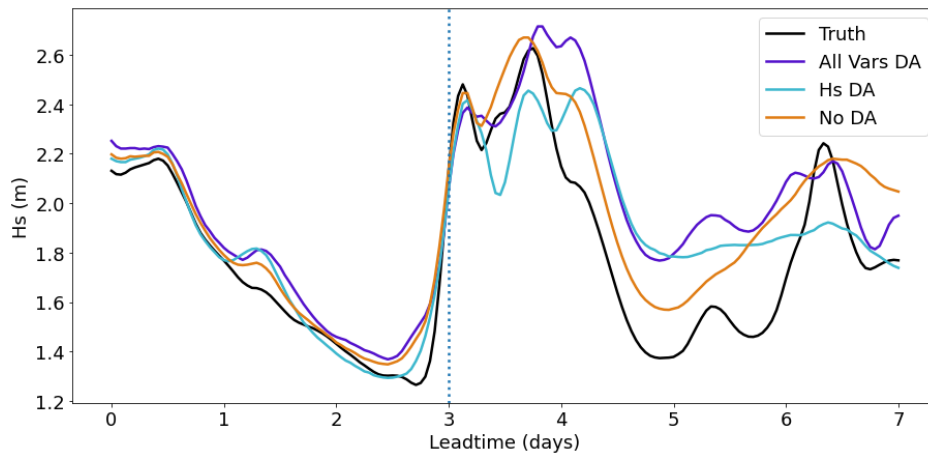


FIGURE 5.2: Timeseries of significant wave height at $30^\circ \text{N } 80^\circ \text{W}$ from 12Z 09/03/2022 model run. Truth is shown in black, and all vars DA, H_s DA and no DA ensemble means are shown in dark blue, light blue and orange respectively. Blue dotted line indicates the end of the assimilation window.

The increments (difference between DA and no DA) in figure 5.2 are quite large. This indicates that there are some large covariances between this point and the buoy, which lines up with the findings of section 4.1. As discussed in that section, these long-distance covariances seem to be somewhat spurious, though further research is needed

to determine to exactly what extent.

It can also be seen that the different forecasts diverge greatly after the swell event at 3-4 days leadtime. This suggests that the DA system has successfully captured the dynamical chaos of the physical wave system.

5.1 Increments

The increments $\mathbf{d}_b^a = \mathbf{x}^a - \mathbf{x}_b$ are the corrections applied to the background state by the DA system (ensemble means). From equation 2.15, the state that minimises the cost function will be close to the background when $\mathbf{P}^f$ is small compared to $\mathbf{R}$. This means that we expect the increments to be small in this situation. We have seen that generally $\mathbf{P}^f$ is small at locations far from the buoy, with some exceptions noted.

From equation 2.16, the increments are a linear function of the innovations $\mathbf{d}_b^o = \mathbf{y} - \mathbf{H}\mathbf{x}_b$. This means that the increments at the buoy location should have the same sign as the innovations, since the forecast error variances $\mathbf{P}_{ii}$ are always positive. This is not necessarily true away from the buoy since the covariance between different locations may be negative. Furthermore, innovations at locations away from the buoy are unavailable since observations were only assimilated at the buoy.

However, due to the twin experiment method, the truth is available at all locations, so the increments can be compared to the negative forecast error $\mathbf{d}_b^t = \mathbf{x}_{true} - \mathbf{x}_b$. It is desirable for the increment to have the same sign as $\mathbf{d}_b^t$, since this means that the DA has moved the analysis state closer to the truth than the forecast state.

Figure 5.3 shows the increments in H_s ensemble mean at 12Z 12/03/2022 taken from the 12Z 10/03/2022 forecast run. All vars DA has larger increments overall, and this is especially true far from the buoy. The increments from both DA runs match the sign of $\mathbf{d}_b^t$ somewhat well, but both have regions where the sign is incorrect.

A notable example is the strong positive increment from all vars DA in the northern GoM, where $\mathbf{d}_b^t$ is slightly negative. The lower magnitude of increments in H_s DA, especially near the buoy, means it undercorrects in some regions where the magnitude of $\mathbf{d}_b^t$ is large, such as just northwest of the buoy in the western GoM. All vars DA does a better job at correcting for this large error.

Figure 5.4 shows the same as 5.3 but for the increments and negative errors in T_{m01} . The equivalent plots of T_{m02} are shown in figure A.1. All vars DA again show larger increments overall, and here all vars DA does a better job of matching the sign of innovation across the wider domain. It captures the strip of negative $\mathbf{d}_b^t$ seen extending through the central GoM, which is missed by H_s DA. However just East of the buoy, H_s DA has positive increments matching $\mathbf{d}_b^t$, while all vars DA is strongly negative.

Overall, all vars DA has generally larger increments across the whole domain and especially at locations far from the buoy. These large, long-distance increments probably

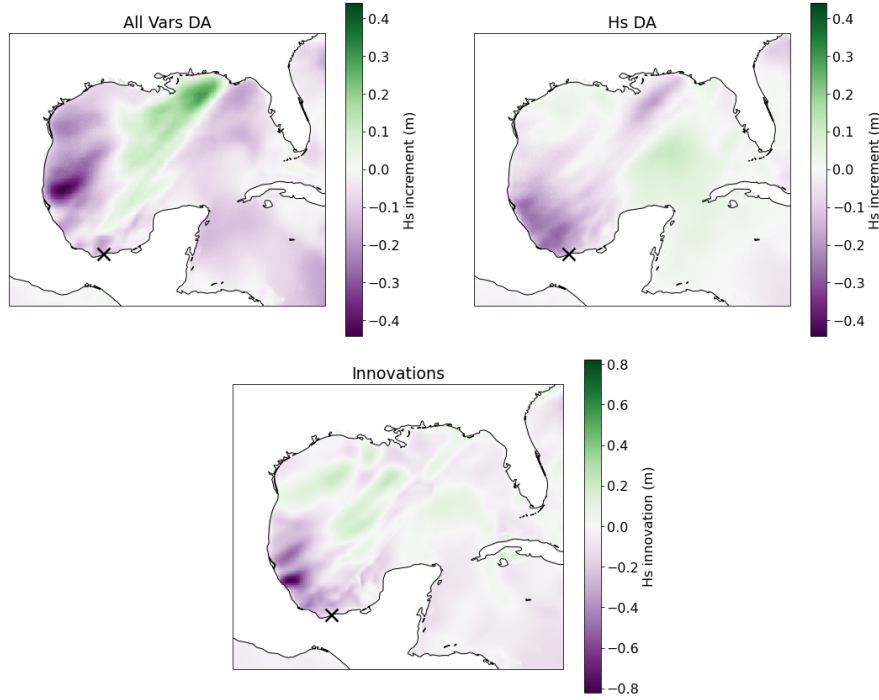


FIGURE 5.3: Significant wave height ensemble mean increments at 12Z 12/03/2022 from all vars DA (top left) and H_s DA (top right), and significant wave height negative errors from the same time (bottom). Taken from 12Z 10/03/2022 forecast run. The buoy location is marked with a cross.

arise from the long distance covariances seen in section 4.1. In that section, we concluded that some of the long distance covariances probably arise spuriously from sampling error, so these increments may be somewhat physically unrealistic. They could also be the result of long-distance correlations in the atmosphere that affect the wave model via forcing. Both DA experiments have increments which largely match the sign of the negative error $\mathbf{d}_b^t$, with all vars DA doing better overall in both variables.

5.2 RMSE

The root mean square error (RMSE) is a very important measure of the error in a forecast. It tells us how much the forecast value diverges from the truth over a given region or time window, and is overall a good measure for the accuracy of a forecast. Here we present the RMSE of the forecast as a function of both leadtime and space, and analyse it in the context of our knowledge of the DA increments and forecast error covariances.

Figure 5.5 shows the RMSE of the ensemble mean from no DA, H_s DA and all vars DA. The equivalent plot of T_{m02} is shown in figure A.2. The mean is taken over a 24-hour period from 12Z 12/03/2022 to 12Z 13/03/2022. This period was chosen because it contains the highest waves in the forecast period, which are associated with higher errors. Forecasts of this period are available from multiple start dates, giving a range of leadtimes. Errors are plotted for H_s and T_{m01} .

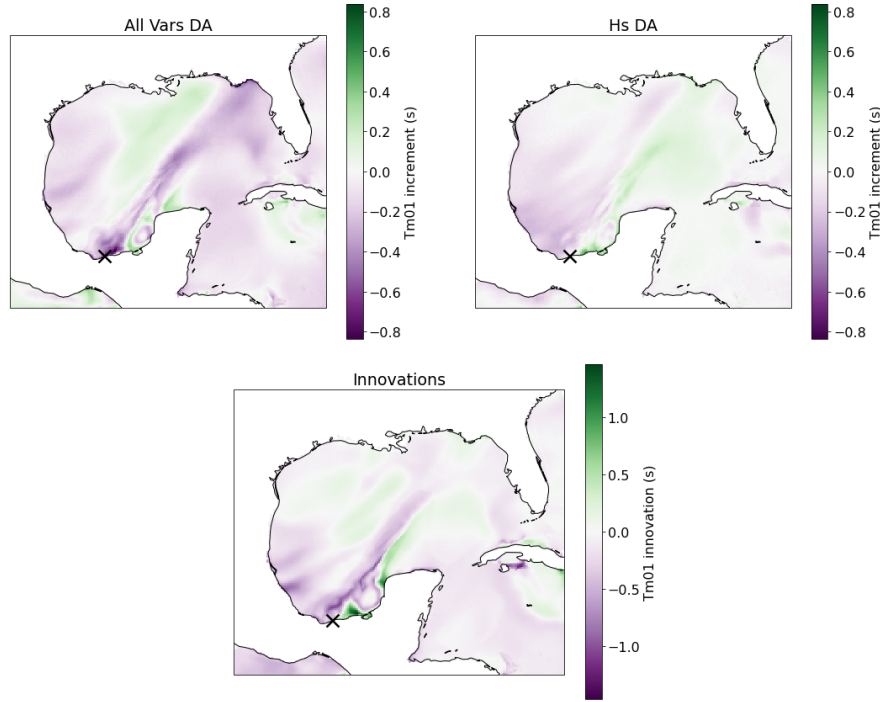


FIGURE 5.4: T_{m01} ensemble mean increments at 12Z 12/03/2022 from all vars DA (top left) and H_s DA (top right), and T_{m01} negative errors from the same time (bottom). Taken from 12Z 10/03/2022 forecast run. The buoy location is marked with a cross.

For H_s , during the assimilation window both H_s DA and all vars DA show an improvement of $\sim 60\%$ over no DA. After the end of the window, the errors from all runs increase and at 3-4 days leadtime, H_s DA shows a much greater RMSE reduction than all vars DA. At 4-5 days, H_s DA has similar error to no DA and all vars DA shows an improvement of $\sim 50\%$. 6-7 days is similar except the improvement from all vars DA is around 35 %. At 5-6 days leadtime, both H_s DA and all vars DA have higher RMSE than no DA.

For T_{m01} the results are similar, with consistent improvement from DA within the assimilation window and more variable results afterwards. All vars DA still performs better than H_s DA with the exception of 1-2 days and 3-4 days leadtime. It is notable that H_s DA gives improvements in this case since no observations of T_{m01} are actually assimilated. This can be explained by the nonzero cross-covariance between H_s and T_{m01} (figure 4.6). One difference between the errors in T_{m01} and H_s is that H_s DA does slightly worse in predicting T_{m01} , when using the other two runs as a benchmark. This is unsurprising as we would expect assimilating H_s , T_{m01} and T_{m02} to give better predictions of T_{m01} than H_s alone.

Overall the performance of both DA experiments is similar during the assimilation window, and after the window all vars DA gives smaller RMSE than H_s DA.

Figure 5.6 shows the spatial distribution of RMSE in H_s ensemble mean over the period from 12Z 12/03/2022 to 12Z 13/03/2022. The middle panels show the RMSE from

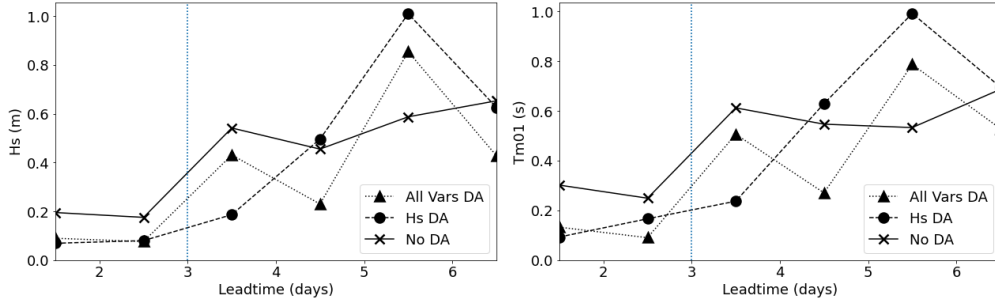


FIGURE 5.5: RMSE in significant wave height (left) and T_{m01} (right) analysis ensemble mean at the buoy location as a function of leadtime. RMSE is calculated over a 24hr period. Results shown for no DA (solid line), H_s DA (dashed line) and all vars DA (black dotted line). Blue dotted line indicates the end of the assimilation window.

no DA, and the side panels show the RMSE reduction achieved by all vars DA, i.e. $RMSE_{No\ DA} - RMSE_{DA}$. So when this quantity is positive, shown by blue regions on the plot, the RMSE with DA is smaller than without.

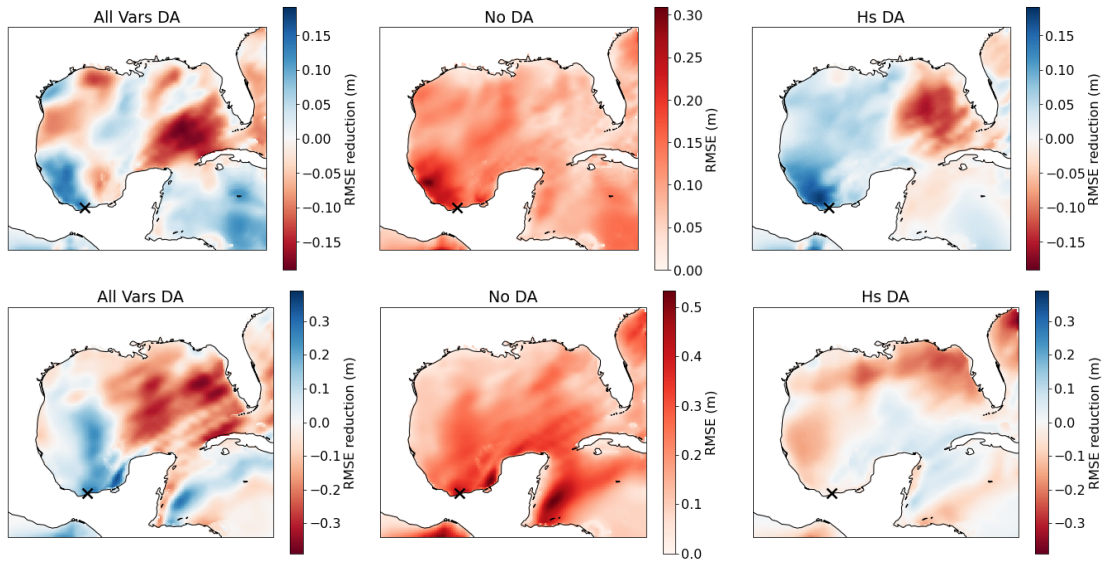


FIGURE 5.6: Maps of RMSE in significant wave height ensemble mean from the period 12Z 12/03/2022 to 13/03/2022, with leadtime 2-3 days (top) and 4-5 days (bottom). Middle panels show RMSE from no DA, side panels show absolute RMSE reduction compared to no DA from all vars DA (left) and H_s DA (right). The buoy location is marked with a cross.

At 2-3 days leadtime, both H_s DA and all vars DA show significant improvements in the immediate vicinity of the buoy. H_s DA shows RMSE reduction in the western GoM that is weaker than the reduction near the buoy, and the eastern GoM has increased RMSE. Outside the GoM there is very little change. All vars DA has more regions within the GoM where the RMSE is increased, and the increase in the eastern GoM is larger than that shown by H_s DA. There is also a noticeable reduction outside the GoM.

At 4-5 days leadtime, all vars DA shows reduced RMSE in the western GoM and increased RMSE further from the buoy in the eastern GoM. The change in RMSE outside the GoM is smaller relative to that seen within the GoM, but the magnitude of RMSE change over the whole domain is generally larger than that seen at 2-3 days. This is probably due in part to the fact that the no DA RMSE is larger. H_s DA shows smaller RMSE changes overall than all vars DA, and there is no significant reduction near the buoy.

Figure 5.7 shows the spatial distribution of RMSE in T_{m01} ensemble mean over the period from 12Z 12/03/2022 to 12Z 13/03/2022. The equivalent plots of T_{m02} are shown in figure A.3. At 2-3 days leadtime, both H_s DA and all vars DA show improvements over no DA in the vicinity of the buoy with some increased errors in the eastern GoM and in the Atlantic. The general scale of error reduction/increase is the same in both cases.

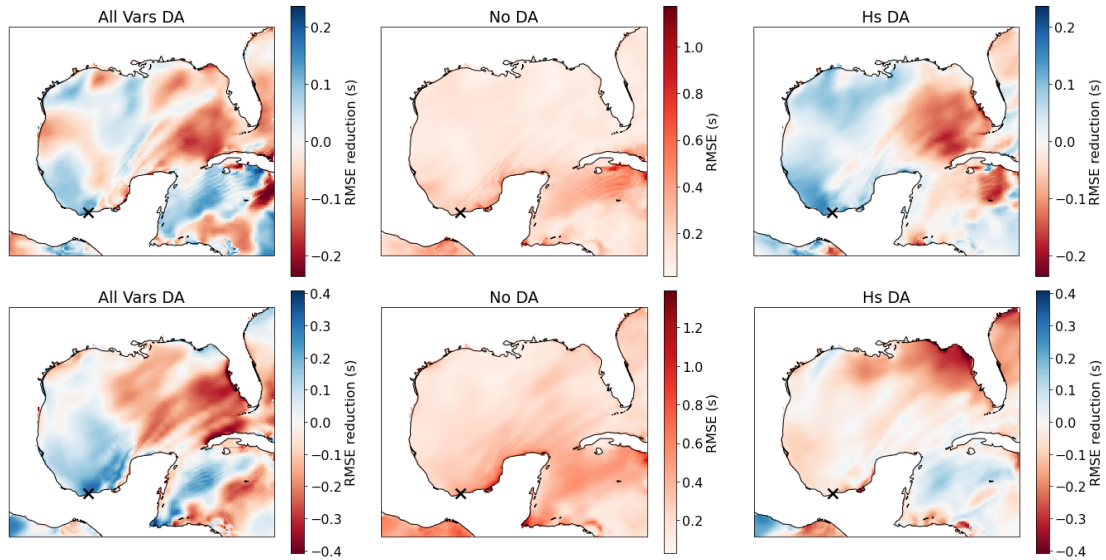


FIGURE 5.7: Maps of RMSE in T_{m01} ensemble mean from the period 12Z 12/03/2022 to 13/03/2022, with leadtime 2-3 days (top) and 4-5 days (bottom). Middle panels show RMSE from no DA, side panels show absolute RMSE reduction compared to no DA from all vars DA (left) and H_s DA (right). The buoy location is marked with a cross.

All vars DA also has regions of increased errors in the western GoM. At 4-5 days leadtime, all vars DA shows improvement through most of the western GoM, with some increased errors again in the eastern GoM and in the Atlantic. H_s DA shows little change in RMSE through most of the domain, with the exception of a large increase in the northeastern GoM. The overall pattern of error reductions is similar to that seen in figure 5.6.

Overall, all vars DA does better than H_s DA at predicting both H_s and T_{m01} near the buoy after the assimilation window, while both perform similarly during the window. The improvements seen during the window are more widespread in H_s DA, while after the end of the window the opposite is true. All vars DA shows larger changes

in errors in general, and particularly at locations away from the buoy, which is in line with the larger increments seen in all vars DA.

5.3 Ensemble Spread

Ensemble spread is an important measure of uncertainty in ensemble forecasts. One aim of this project is to reduce uncertainty in the forecasts so that they may be used to inform decisions with greater confidence. Figure 5.8 shows timeseries of ensemble standard deviation in H_s and T_{m01} at the buoy location from two different model runs. The equivalent plots of T_{m02} are shown in figure A.4. Both H_s DA and all vars DA show a reduction in ensemble spread in both variables, as is expected from DA (section 2.2.3).

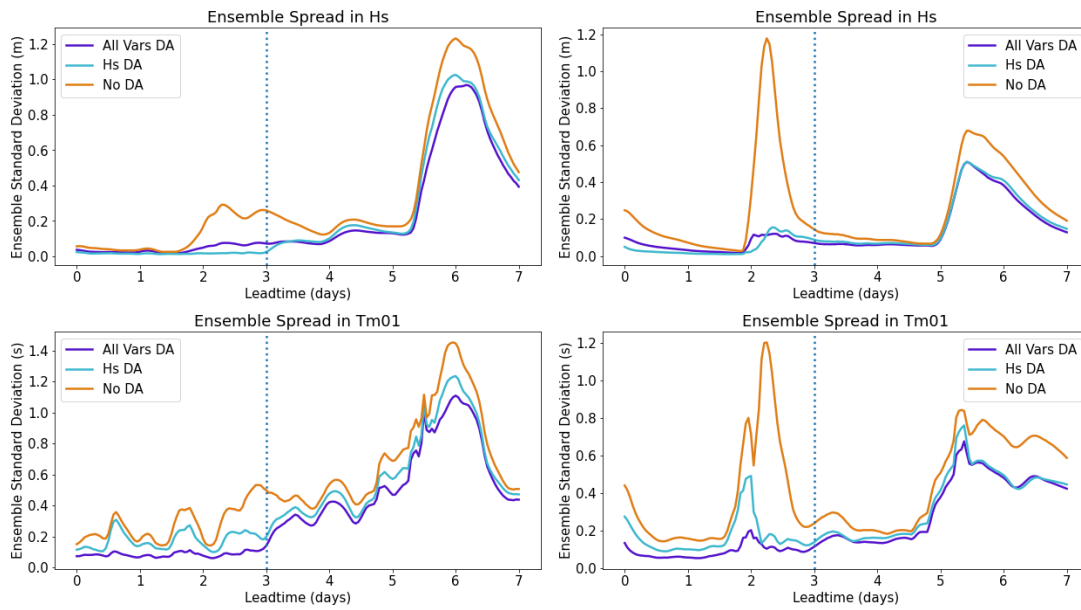


FIGURE 5.8: Timeseries of ensemble standard deviation in significant wave height (top) and T_{m01} (bottom) at the buoy location for 12Z 06/03/2022 (left) and 12Z 10/03/2022 (right) model runs.

All vars DA shows a greater reduction in T_{m01} ensemble spread overall. This is unsurprising since T_{m01} is not directly assimilated in H_s DA. In H_s ensemble spread, both runs perform similarly overall, but H_s DA generally shows lower spread during the assimilation window and the opposite is true after the assimilation window. This is likely because the observational error scale in H_s was increased from 10% in H_s DA to 12% in all vars DA, thus reducing the effect of the assimilation.

Figure 5.9 shows the spatial distribution of H_s ensemble spread at 12Z 11/03/2022, from 1 day and 5 days leadtime. At both leadtimes, All vars DA shows a greater reduction in ensemble spread than H_s DA across most of the domain. The only exception to this is the vicinity of the buoy at 1 day leadtime, when H_s DA shows a greater reduction. This is similar to what was seen in the timeseries, when H_s DA had greater spread reduction outside of the assimilation window. This reflects the fact that the entire 7-day forecast window is updated in one step by the DA system, so the dimensions of

space and time within a forecast window are treated equivalently. With this in mind, it is not surprising to see parallels in the spatial and temporal distributions of ensemble spread. The greatest reductions in spread are where the spread of no DA is greatest.

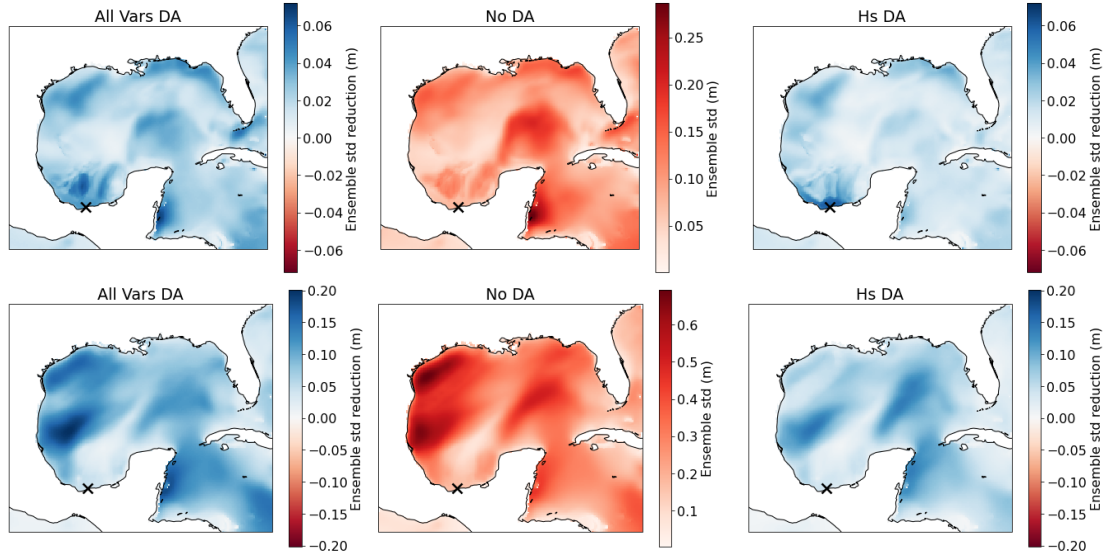


FIGURE 5.9: Ensemble standard deviation in significant wave height at 12Z 11/03/2022, taken from 12Z 10/03/2022 (1 day leadtime, top) and 12Z 06/03/2022 (5 days leadtime, bottom) model runs. Middle panel shows standard deviation from no DA, side panels show standard deviation reduction from all vars DA (left) and H_s DA (right).

Figure 5.10 shows the same as 5.9 but for T_{m01} . The equivalent plots of T_{m02} are shown in figure A.5. At 1 day leadtime, the pattern of reductions from both all vars DA and H_s DA closely match the pattern of standard deviation from no DA, and the reductions from all vars DA are stronger overall. At 5 days leadtime, the reduction from all vars DA generally matches the pattern of standard deviation from no DA, with a peak concentrated on the western side of the Yucatan peninsula. This peak is also present in H_s DA, which shows little reduction in spread elsewhere.

Overall, all vars DA shows greater reduction in ensemble spread at most times and locations, except near the buoy during the assimilation window. The reductions in spread are greatest where the no DA ensemble spread is greatest, which is good as this reduces the maximum uncertainty in the forecast.

It is not desirable to focus solely on reducing the ensemble spread, as this may lead to the ensemble being overconfident. As described in section 2.2.3, it is detrimental to the forecast to have ensemble spread smaller than the forecast errors as this can lead to too much confidence being placed in an inaccurate forecast. An example is the strong reduction in spread seen in the eastern GoM, where RMSE generally increases. To test the prevalence of this problem, the ensemble standard deviation is plotted against the RMSE.

Figure 5.11 shows a scatter plot of ensemble standard deviation against RMSE at the

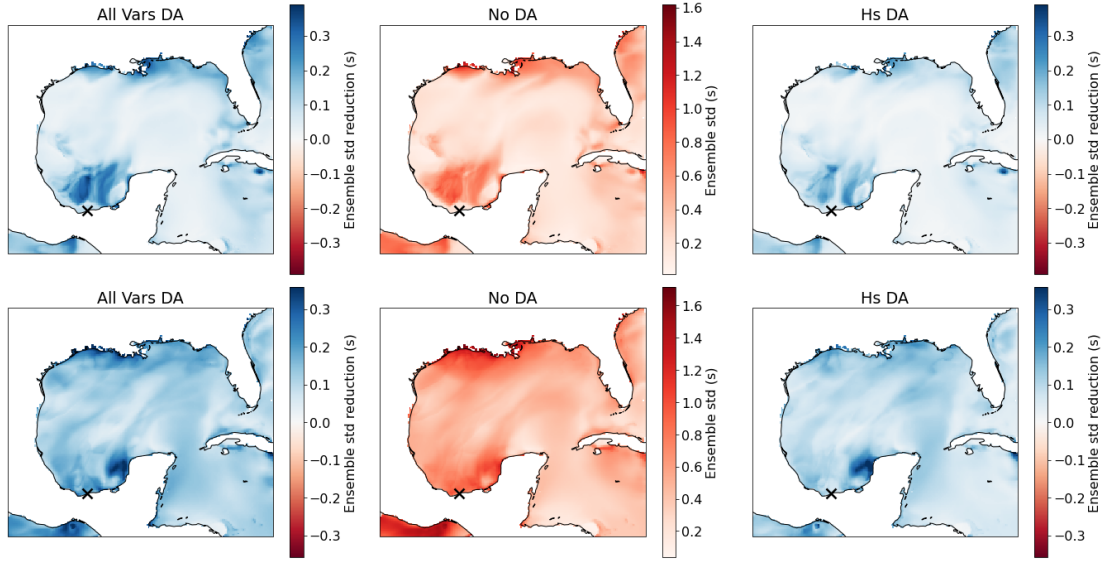


FIGURE 5.10: Ensemble standard deviation in T_{m01} at 12Z 11/03/2022, taken from 12Z 10/03/2022 (1 day leadtime, top) and 12Z 06/03/2022 (5 days leadtime, bottom) model runs. Middle panel shows standard deviation from no DA, side panels show standard deviation reduction from all vars DA (left) and H_s DA (right).

No RMSE Threshold			With RMSE Threshold		
	H_s freq.	T_{m01} freq.		H_s freq.	T_{m01} freq.
No DA	21	19	No DA	8	18
H_s DA	291	250	H_s DA	115	214
All Vars DA	212	293	All Vars DA	100	135

TABLE 5.1: Left: Frequency of overconfident significant wave height and T_{m01} ensembles from no DA, H_s DA and all vars DA. Right: The same but the ensemble must meet a minimum RMSE threshold of 0.1m for significant wave height and 0.2s for T_{m01} to be counted.

buoy location, for H_s and T_{m01} . The equivalent plot of T_{m02} is shown in figure A.6. RMSE is calculated from the ensemble mean and averaged over a 6 hour rolling window in leadtime. Data from all forecast windows from H_s DA and all vars DA is included, with H_s DA represented by circles and all vars DA represented by triangles. The leadtime of each data point within its forecast window is shown by its colour. Lines of best fit are calculated for both H_s DA and all vars DA, and plotted along with the 1:1 line.

The lines of best fit show that both DA runs have RMSE generally below the ensemble spread, with all vars DA being lower than H_s DA overall. This is desirable as it shows that the DA has not decreased ensemble spread to the point where it is no longer an appropriate measure of uncertainty. Data points above the 1:1 line correspond to overconfident ensembles, so for both variables from both DA runs, the number of data points above the line was counted and shown in table 5.1. The equivalent counts from no DA are also included.

The total number of data points in each run is 984, so all vars DA gives overconfident

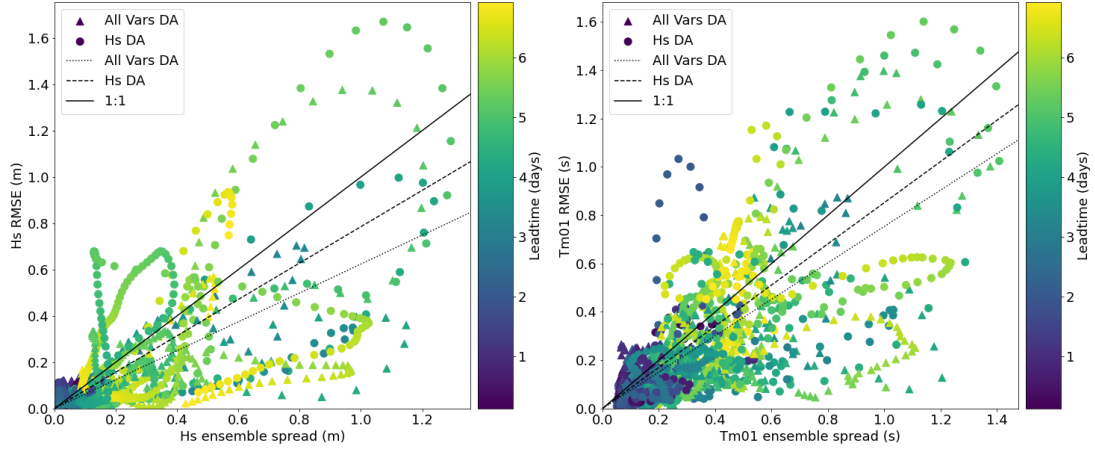


FIGURE 5.11: Left: Scatter plot of significant wave height RMSE against significant wave height ensemble standard deviation at the buoy location. All vars DA shown with triangles, H_s DA shown with circles and leadtime shown by colour. Lines of best fit shown for all vars DA (dotted) and H_s DA (dashed) as well as 1:1 line (solid). Right: the same for RMSE and ensemble spread in T_{m01} .

H_s ensembles about 22% of the time and overconfident T_{m01} ensembles about 30% of the time. H_s DA gives overconfident H_s ensembles about 30% of the time and overconfident T_{m01} ensembles about 25% of the time. These are both much greater than in the forecast prior to DA. The colours on the figure show that most of the occasions where RMSE is significantly more than the spread are late in the forecast window (leadtime > 4 days) with few exceptions.

So H_s DA gives more overconfident H_s ensembles and all vars DA gives more overconfident T_{m01} ensembles, with H_s DA giving more overconfident ensembles overall. The high prevalence of overconfident T_{m01} ensembles from all vars DA lines up with the decreased spread in T_{m01} from all vars DA seen in figure 5.8, and may be due to the increased reduction in ensemble spread from multiple variables being assimilated.

Overconfident ensembles are most important to avoid when the error is large, since larger forecast errors present greater risk for incorrect decision making. So the count of the overconfident ensembles was repeated with a minimum threshold that the RMSE must meet in order for the ensemble to be counted. These thresholds are 0.1 m for H_s and 0.2 s for T_{m01} , and were set at values that judged to be acceptable levels of error in each variable. After applying these thresholds, the frequency of overconfident ensembles reduced by at least 50% in all cases except in the T_{m01} ensembles from H_s DA. This reflects the fact that the errors in T_{m01} from H_s DA are large. This now gives overconfidence prevalences of 10-20% in most cases.

In summary, the DA has reduced the ensemble spread as expected, with all vars DA having a stronger effect than H_s DA. The greatest reductions in spread occur where the pre-DA spread is greatest. Both DA experiments increase the prevalence of overconfident ensembles, but the ensemble spread is still less than the RMSE in general, so it is

judged to still be a good measure of uncertainty.

5.4 Errors in Maximum H_s

H_s maxima are of particular interest in this project as they present the most danger to offshore operations. In this section we examine the effect of DA on the errors in the timing and magnitude of the H_s maximum on 12/03/2022. Figure 5.12 shows scatter plots of the timing and magnitude from no DA, H_s DA and all vars DA, relative to the truth. The maximum is selected from the period spanning 12Z 12/03/2022 to 12Z 13/03/2022. The left plot is generated from the model run starting 12Z 10/03/2022 and the right plot is generated from the model run starting 12Z 08/03/2022, giving leadtime ranges of 2-3 days and 4-5 days respectively. Table 5.2 shows the ensemble RMSE in timing and magnitude from each ensemble for both runs.

Note on the scatter plots that the points are arranged discretely at integer numbers of hours: this is simply because the forecast data used has a time resolution of one hour. This could reduce the maximum significant wave height detected if the true maximum lies in the middle of two timesteps. One way around this may be to fit a quadratic polynomial to the model data points either side of the peak and find the maximum value of this polynomial, but this was not explored here. The forecasters at Fugro work regularly with hourly forecast data without any interpolation applied, and we assume they are aware that this can lead to inaccurate maximum values being sampled.

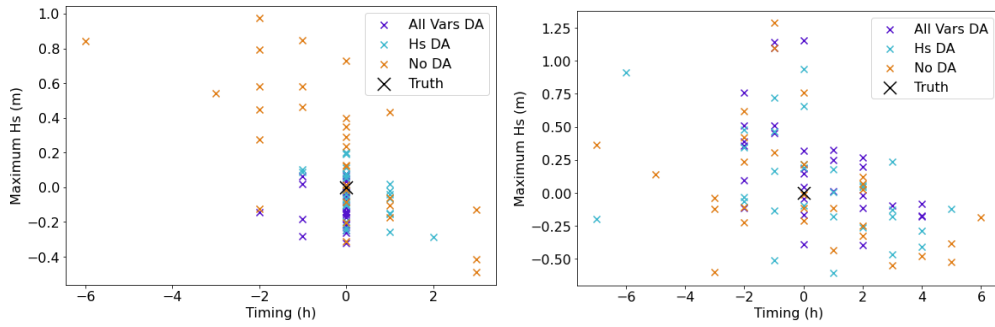


FIGURE 5.12: Scatter plots of timing and magnitude of maximum significant wave height at buoy location from period 12Z 12/03/2022 to 12Z 13/03/2022. Generated from model run starting 12Z 10/03/2022 (left) and 12Z 08/03/2022 (right). Truth is shown in black, and the no DA, H_s DA and all vars DA ensemble members are shown in red, blue and green respectively.

Leadtime 2-3 days			Leadtime 4-5 days		
	Timing (h)	Magnitude (m)		Timing (h)	Magnitude (m)
No DA	1.9	0.46	No DA	2.9	0.45
H_s DA	0.66	0.14	H_s DA	2.7	0.40
All Vars DA	0.51	0.15	All Vars DA	1.9	0.46

TABLE 5.2: RMSE of magnitude and timing of the significant wave height maximum from the ensembles plotted in figure 5.12.

At a leadtime of 2-3 days, there is a large improvement in forecasts of timing and magnitude from both DA runs. From the scatter plot, the ensemble spread is considerably tighter around the true value, and the RMSE in both quantities is reduced. All vars DA does slightly better than H_s DA in predicting the timing of the maximum, and vice versa for the magnitude, but the differences between the two are small compared to the difference of both with no DA.

After 4-5 days, the errors are larger in all quantities but the magnitude from the no DA run. The differences between the two DA runs is larger, as H_s DA has a similar error in timing to no DA, but shows a small improvement in the magnitude of 13%. Conversely, all vars DA has similar error in magnitude to no DA, but shows an improvement in timing of 35%.

Overall, H_s DA and all vars DA perform similarly to each other during the assimilation window, and both significantly outperform no DA as expected during this time. As seen in other aspects, their performance differs more after the assimilation window, with H_s DA providing better forecasts of the magnitude of the peak and all vars DA providing better forecasts of its timing.

5.5 Chapter Summary

In this chapter, the results of the DA experiments have been presented and explained with reference to the theory from chapter 2 and the errors seen in chapter 4. The aspects of the results examined are the increments, RMSE, ensemble spread and errors in H_s maximum. All vars DA is seen to outperform H_s DA in most aspects, with some exceptions being the H_s RMSE across the western GoM during the assimilation window and the magnitude of the H_s maximum on 12/03/2022.

Chapter 6

Conclusions

6.1 Discussion and Conclusions

The main aim of this study was to determine what improvements can be made to Fugro's Gulf of Mexico wave forecasts by assimilating the observations available to Fugro using DA. To that end, observations of H_s , T_{m01} and T_{m02} were assimilated using a method which borrows from the ensemble transform Kalman filter. Various aspects of the forecasts before and after DA were compared to evaluate the success of the DA. Particular attention was paid to the large swell event on 12/03/2022, as events of this kind pose large hazards to offshore operations.

Prior to applying DA, the background and observational errors were examined, as these are crucial to determining the size and structure of the DA increments. Investigating the background errors, which are determined from the ensemble, it was found that the ensemble variance in H_s is a strong function of H_s itself, and it peaks just before the peak in H_s . A similar relationship was found for T_{m01} . We saw that the forecast error covariances are often strongest near the buoy location, especially to the northwest. Longer distance covariances are generally weaker, but can arise physically due to winds and travelling waves and also spuriously due to sampling error.

The observation errors were first examined using a collocation study. The observation errors were seen to be a strong linear function of the variable being observed, particularly in H_s . This motivated setting an observation error standard deviation in the DA system which depends linearly on the observations. Since the scaling factor for the periods was small, a constant minimum error threshold was implemented for these variables to ensure that the error would not become too small and negatively impact DA performance. This threshold was estimated following Desroziers et al. (2005).

We compared the results of assimilating H_s alone (H_s DA) to assimilating all three variables (all vars DA), and found that both approaches, but particularly all vars DA, produced possibly unrealistic increments with large changes to the state far away from the observation location. Comparing our results to Houghton et al. (2023) (figure 1.3 and figure 5.3) shows that their LETKF method produced more realistic increments, which captured the flow dependency of the background covariances while keeping

the effects of the observations confined to the vicinity of the observation location. This is because the LETKF method enforces shorter correlation length scales via localisation.

The RMSE was examined as this is an important diagnostic of forecast skill. DA gave reductions in RMSE at the buoy of up to 50% in all variables, which extended up to 0.5 days beyond the end of the assimilation window in the case of H_s DA, and 1.5 days beyond in the case of all vars DA. Both cases produced very similar RMSE reductions during the assimilation window, and all vars DA had consistently smaller errors than H_s DA at >1 day after the end of assimilation. At 2-3 days after assimilation, both DA runs had larger errors than the no DA case. Looking at the wider domain, we found that H_s DA showed more consistent and widespread RMSE reductions during the assimilation window, and the opposite is true after the window.

These improvements in RMSE are less consistent and slightly shorter lived than those obtained in Houghton et al. (2022, 2023) (figures 1.3, 1.4). This reflects the two major limitations identified in section 1.2, namely the limited number of observations available to us and the fact that the forecast model is unavailable, which necessitates updating an entire forecast run in one update step and not use cycling. Our results agree with Houghton et al. (2022) in that assimilating multiple variables (spectral DA) performed better than assimilating wave height alone in most cases. One difference between the results of that study and this is that Houghton et al. (2022) found H_s DA and spectral DA to be virtually identical in predicting H_s , while spectral DA was better for predicting other variables. We found that all vars DA did better than H_s DA for all the variables tested. This may reflect the choice to assimilate a selection of bulk variables rather than the full spectrum.

The RMSE improvements seen in this study are more similar to those in Caires et al. (2018), which likely reflects the smaller number of observations used in that study. The fact that we achieved similar results to Caires et al. (2018) despite not being able to run the forecast model may indicate that the limited number of observations is the more important limitation out of the two. It may also be due to other differences in methodology between this study and that, such as a different DA method, compensating for the lack of a forecast model in this study.

The ensemble spread was examined as this is a commonly used measure of forecast uncertainty. We found that the ensemble spread is reduced by both all vars DA and H_s DA. For H_s forecasts, H_s DA showed greater reductions in spread near the buoy during the assimilation window and all vars DA showed greater reductions at other locations and times. We found that post DA, there was some slight degradation in the suitability of ensemble spread as a measure of forecast uncertainty. There were more instances of what we call ‘overconfident’ ensembles, where there is a large error which the ensemble spread does not capture, which may pose a risk if the forecasts are used to inform decision making in offshore operations. However, it was seen that in general the ensemble spread is still less than the forecast RMSE so it is judged that ensemble

spread is still a good measure of forecast uncertainty. This aspect of performance could be improved with inflation and/or further tuning of observation error scales, and other improvements to the DA method such as the introduction of cycling could reduce error overall, which would also help in this regard.

DA was found to reduce the errors in forecasts of the H_s maximum associated with the cold front on 12/03/2022. With a 2-3 day leadtime (so the maximum was during the assimilation window), both H_s DA and all vars DA showed improvements showed improvements of around 65% in forecasts of both the timing and magnitude of the H_s maximum. At 4-5 days leadtime, when the maximum was outside of the assimilation window, all vars DA showed improvement in forecasting the timing of the peak, while H_s DA did better forecasting its magnitude.

Overall, improvements were seen from DA in all aspects of the forecasts. All vars DA was seen to generally outperform H_s DA, with some noted exceptions such as the magnitude of the H_s maximum on 12/03/2022. These results are promising given the simplicity of the DA implementation and the limitations noted, and we expect that future studies using more refined methods will be able to achieve better results.

6.2 Limitations and Future Work

As noted before, two major limitations of this study are the small number of observations available and the lack of access to a forecast model which makes cycling impossible. The first of these limitations could be mitigated by augmenting the observations used here with other wave observations in the GoM. NOAA maintains a public database of up to date observations from many fixed buoys around the world. Several of these buoys are located in the GoM and measure the same bulk quantities assimilated in this project. A map showing the locations of a selection of these buoys is shown in figure 6.1.

As demonstrated in section 1.2, DA tends to do better when it has more observations to work with, so including data from these buoys may improve the DA performance. It is also worth restating here that the twin experiment methodology used here excluded the possibility of correcting any biases present in the forecast data, so future work assimilating real observations would be useful to investigate the effectiveness of DA for this purpose and more robustly evaluate the utility of the buoy observations.

As well as having limited observations, we also had limited access to forecast data, with only seven overlapping week-long forecast windows available. This raises questions over the robustness of the DA results, and so it would be useful for future studies to include more forecast data.

The lack of a forecast model will not be an issue in any future operational DA implementation undertaken by Fugro. There are many ways in which the method used in this study could be modified to include cycling, and perhaps one of the simplest

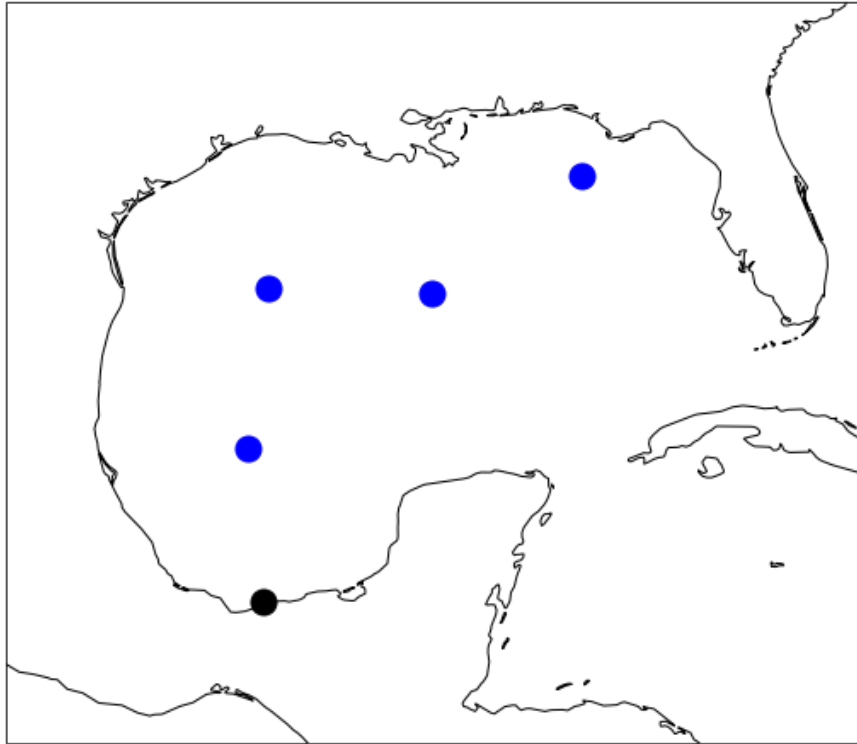


FIGURE 6.1: Locations of buoys with observations of relevant wave variables. The buoy used in this project is marked in black, and other buoys are marked in blue

is to assimilate the observations as they are available in a more conventional ETKF setup. This would reduce the dimension of the state space since the state vector would only contain values from a single timestep. Observations could be assimilated every hour, or some observations could be skipped so that the DA could be performed less frequently to reduce computational cost. We have seen in this study that the ETKF update step excessively reduces the ensemble spread in comparison to the forecast error, so this proposed method would be susceptible to filter divergence and some inflation or other preventative measure would likely be necessary to mitigate this.

Another possible approach would be to use a more conventional smoother method such as 4DEnVar. The method used in this study is a smoother, but the implementation is unconventional and so may be susceptible to unidentified problems. A smoother method simultaneously assimilates observations from an entire window, rather than assimilating observations from each observation time one-by-one as in a filter method. This would allow assimilation to be done less frequently while still assimilating all available observations, and is less prone to filter divergence. However, this method would require the augmentation of the state vector to include values from multiple timesteps, and would require minimisation of the cost function using a method such as the conjugate gradient method (Nocedal and Wright, 2006) which is typically computationally expensive. Further research should be done to determine the best approach. Schematics of both methods outlined here are shown in figure 6.2.

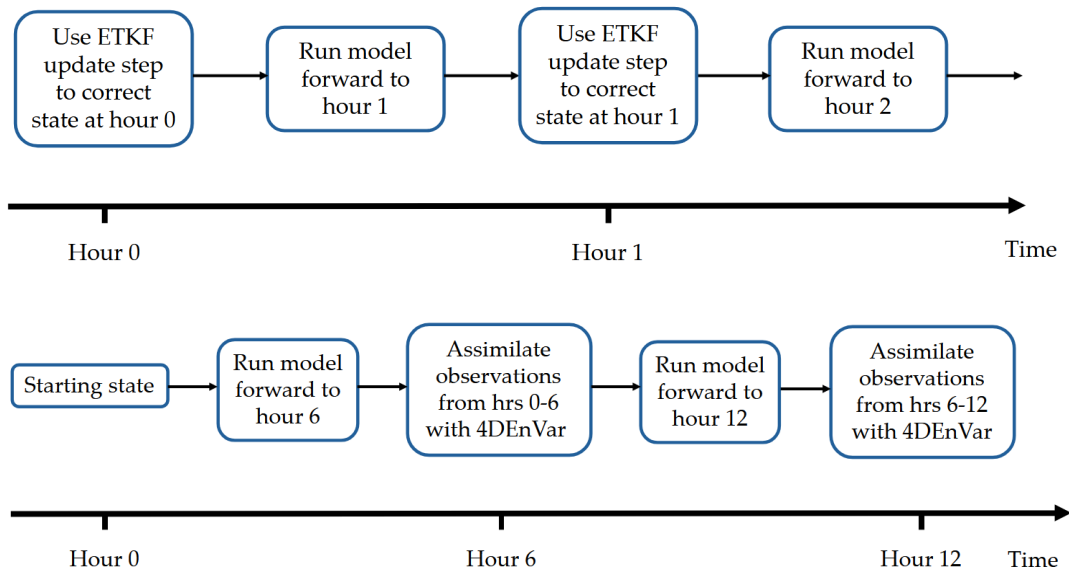


FIGURE 6.2: Schematics of potential DA methods which incorporate cycling. Top: Observations assimilated hourly using a conventional ETKF filter method. Bottom: Observations assimilated every 6 hours using the 4DEnVar smoother method.

Since we have seen that the ensemble spread is reduced too much by the DA method used in this project, future studies into this topic could include some kind of countermeasure against this effect. One possible countermeasure which has been discussed is inflation applied prior to the update step, which involves multiplying $\mathbf{P}^f$ by some predefined factor $\alpha > 1$ before updating the model states. It is also possible to apply inflation during the analysis step by multiplying $\mathbf{P}^a$ by α prior to generating the analysis ensemble. Other potential methods for artificially increasing the ensemble spread are adding noise to the forecast ensemble prior to calculating $\mathbf{P}^f$ (Anderson and Anderson, 1999), or the method of relaxation from Whitaker and Hamill (2012), in which the analysis ensemble is modified to bring its standard deviation closer to that of the forecast ensemble while leaving its mean unaffected.

Another problem encountered in this study is the large covariances and increments seen far away from the observation location. As discussed, this is often physically unrealistic, with spurious long distance covariances in $\mathbf{P}^f$ arising from sampling error and causing unrealistic large long-distance increments (Hunt et al., 2007). This may be improved with localisation, but since at least some of the long-distance covariances seen appeared to be genuine, we did not explore this in this study and left it instead for future work.

To inform the implementation of localisation, it would be useful to evaluate to what extent the long-distance covariances are realistic and to what extent they are spurious. One way to do this would be to repeat the experiment with a larger forecast ensemble. The spurious covariances arise due to sampling error, which has less effect when the

ensemble size is larger (Sacher and Bartello, 2008), so spurious covariances should be smaller with a larger ensemble. Another potential route would be investigating the physical mechanisms behind the covariances. A preliminary attempt at this was made in section 4.1, but a more robust understanding of the mechanisms would be necessary to confidently distinguish realistic covariances from spurious ones and inform the implementation of localisation.

We hope that this study has demonstrated the potential of DA to improve ocean wave forecasts, with positive ramifications for safety in offshore operations. This is a promising avenue for future work, and the limitations and suggestions laid out in this section should provide a useful basis for any such future studies to improve on the results obtained here.

Appendix A

Results - T_{m02}

This appendix contains plots of T_{m02} equivalent to the plots of H_s and T_{m01} shown in chapter 5. As the results for T_{m01} and T_{m02} are so similar, these plots are not included in the main body of the text to aid readability, but are given here instead for completeness.

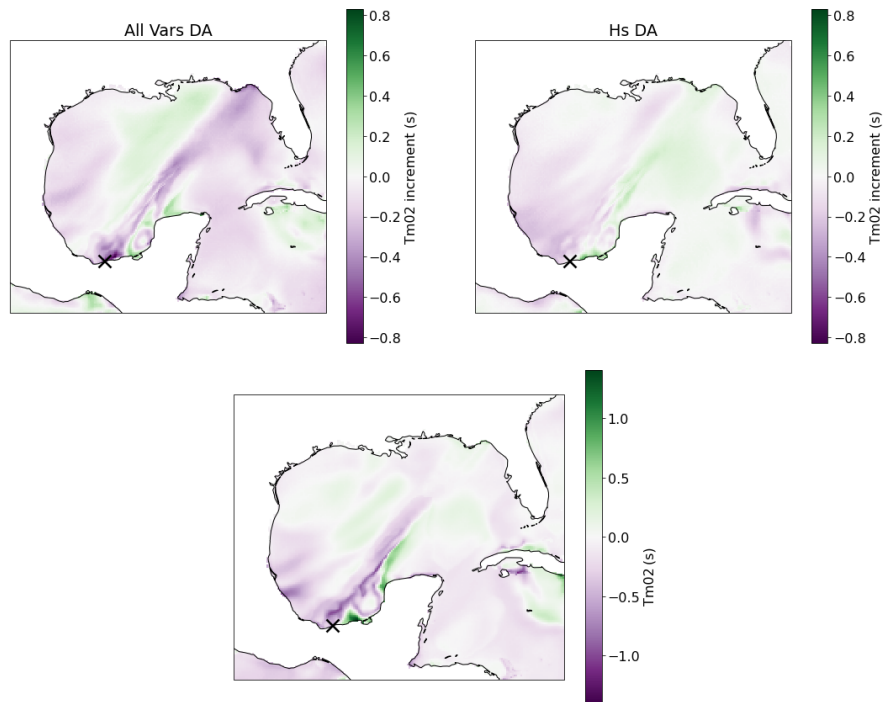


FIGURE A.1: T_{m02} ensemble mean increments at 12Z 12/03/2022 from all vars DA (top left) and H_s DA (top right), and H_s negative errors from the same time (bottom). Taken from 12Z 10/03/2022 forecast run. The buoy location is marked with a cross. Equivalent to figure 5.4.

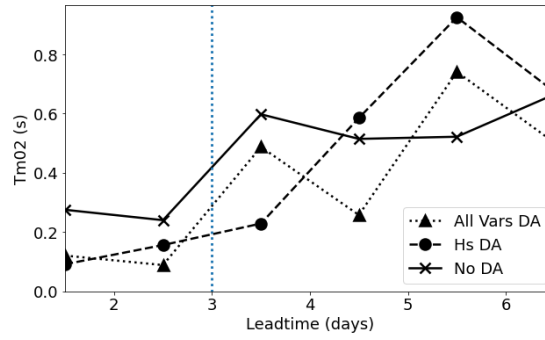


FIGURE A.2: RMSE in T_{m02} ensemble mean at the buoy location as a function of leadtime. RMSE is calculated over a 24hr period. Results shown for no DA (solid line), H_s DA (dashed line) and all vars DA (black dotted line). Blue dotted line indicates the end of the assimilation window. Equivalent to figure 5.5.

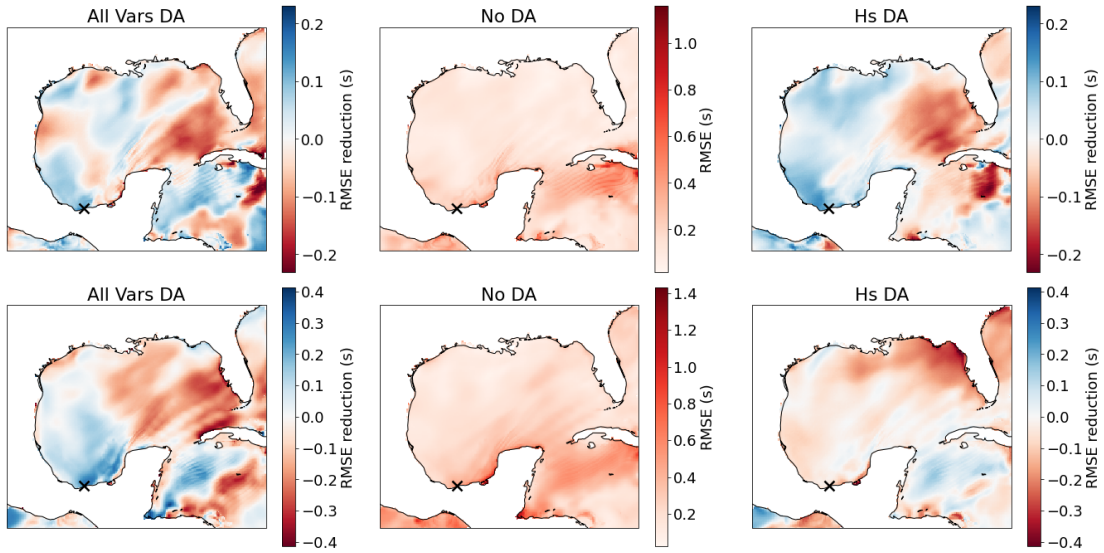


FIGURE A.3: Maps of RMSE in T_{m02} ensemble mean from the period 12Z 12/03/2022 to 13/03/2022, with leadtime 2-3 days (top) and 4-5 days (bottom). Middle panels show RMSE from no DA, side panels show absolute RMSE reduction compared to no DA from all vars DA (left) and H_s DA (right). The buoy location is marked with a cross. Equivalent to figure 5.7.

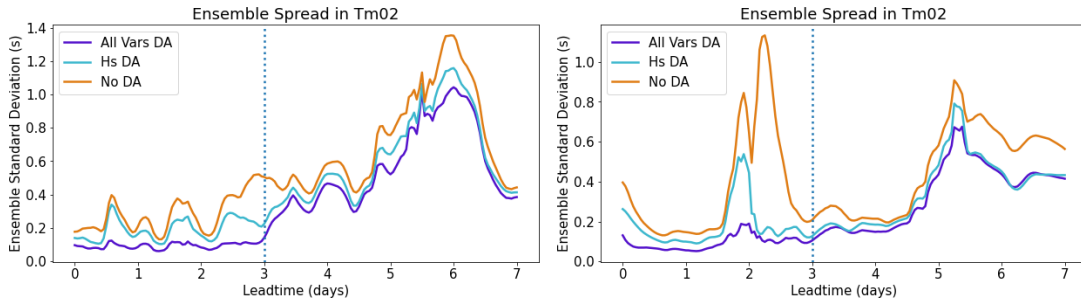


FIGURE A.4: Timeseries of ensemble standard deviation in T_{m02} at the buoy location. Taken from 12Z 06/03/2022 (left) and 12Z 10/03/2022 (right) model runs. Equivalent to figure 5.8.

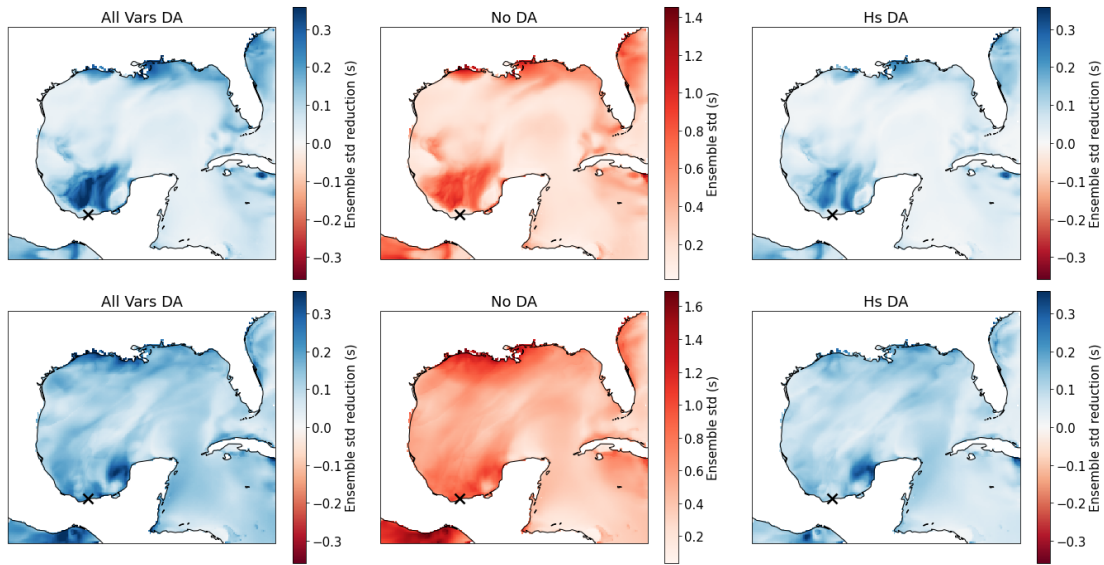


FIGURE A.5: Ensemble standard deviation in T_{m02} at 12Z 11/03/2022, taken from 12Z 10/03/2022 (1 day leadtime, top) and 12Z 06/03/2022 (5 days leadtime, bottom) model runs. Middle panel shows standard deviation from no DA, side panels show standard deviation reduction from all vars DA (left) and H_s DA (right). Equivalent to figure 5.10.

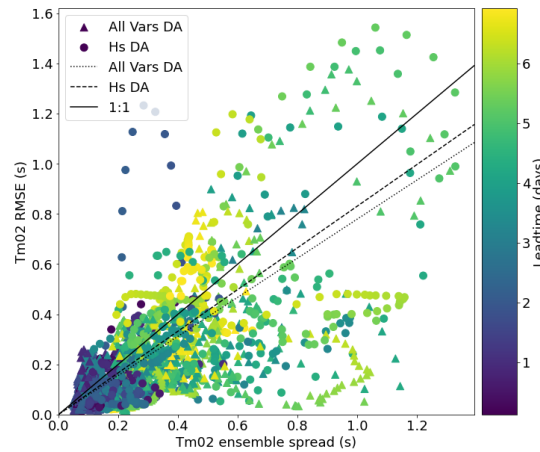


FIGURE A.6: Scatter plot of T_{m02} RMSE against T_{m02} ensemble standard deviation at the buoy location. All vars DA shown with triangles, H_s DA shown with circles and leadtime shown by colour. Lines of best fit shown for all vars DA (dotted) and H_s DA (dashed) as well as 1:1 line (solid). Equivalent to figure 5.11.

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